NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

A STATE UNIVERSITY

UNDER DELHI ACT 06 OF 2018, GOVT. OF NCT OF DELHI

Azad Hind Fauj Marg, Sector-3, Dwarka, New Delhi-110078

SCHEME OF COURSES AND EXAMINATION
FOR
BACHELOR OF TECHNOLOGY
ELECTRONICS & COMMUNICATION

(Effective from the Session: 2019-2020)

APPROVED BY

The Senate in its II to VII meetingS

The Board of Management in its meeting held on ----- --, 20--

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1. INTRODUCTION

NSUT has embarked on its journey towards excellence in academics through the introduction of a novel system of learning that is being followed in many reputed universities globally. The Choice Based Credit System (CBCS) has been proposed by University Grants Commission (UGC) on recommendations of the National Knowledge Commission, to improve the quality of higher education in India. NSUT proposes to adopt CBCS for its Bachelor of Technology courses

CBCS is the mother of student centric educational reforms. A student is provided with an academically rich, highly flexible learning system blended with abundant provision for skill practice and activity orientation that he/she could learn in depth without sacrificing his/her creativity. A student can exercise the option to decide his/her own pace of learning- slow, normal or accelerated plan and sequence his/her choice of paper, learn to face challenges through term work/ project work and may venture out to acquire extra knowledge/ proficiency through add- on facilities. The great advantage of CBCS is that the learning process is made continuous and the evaluation process is not only made continuous but also made learner-centric and is designed to recognize the capability and talent of a student.

2. CURRICULUM STRUCTURE

B.Tech. programme of the University shall be based upon CBCS and shall have well defined Programme Educational Objectives (PEOs). All the courses shall have well-defined Course Outcomes (COs).Courses shall be of three kinds: Core, Elective and Foundation.

a. Core Course (CC): This is a course which is to be compulsorily studied by a student as a core requirement to complete the requirements of the B.Tech. programme.

b. Elective Course: This is a course which can be chosen from a pool of elective courses. It is intended to support the discipline of study by providing an expanded scope, enabling exposure to another discipline/domain and nurturing a student's proficiency and skill. An elective may be of the following types:

i. Discipline Centric Elective (ED): It is an elective course that adds proficiency to the students in the discipline.

ii. Generic Elective (EG): It is an elective course taken from other engineering subjects and enhances the generic proficiency and interdisciplinary perspective of students.

iii. Open Elective (EO): It is an elective course taken from a common pool of non-engineering disciplines that broadens the perspective of an engineering student. These electives shall comprise two groups: Open electives of the Humanities, Social Sciences and Management group and Open electives of the Sciences group.

c. Foundation Course: A Foundation course leads to knowledge enhancement and provides value-based training. Foundation courses may be of two kinds:

i. Compulsory Foundation (FC): It is based upon the content that leads to fundamental knowledge enhancement in Sciences, Humanities, Social Sciences and Basic engineering. They are mandatory for all disciplines.

ii. Elective Foundation (FE): It can be taken from among a common pool of foundation courses which aim at value-based education. They may provide hands-on training to improve competencies, skills or provide education on human, societal, environmental and national values. These shall be mandatory, non-credit courses, which do not carry any credits but a student has to pass in order to be eligible for award of degree.

2.1 EVALUATION AND ASSESSMENT

2.2 SEMESTER WISE COURSE/CREDIT DISTRIBUTION

Table 3 :

Semester	Types of courses as per NSUT Nomenclature						Courses/credits	Credits
	FE (NON-CREDIT)	FC	CC	ED	ED/EG/EO/ EO- Sciences / EO-SS & Mgmt	Training Project etc.		
I	01	05	00	00	00	00	06 courses 20 credits	84 credits
II	00	03	03	00	00	00	06 courses 24 credits	
III	01	00	05	00	00	00	06 courses 20 credits	
IV	01	00	05	00	00	00	06 courses 20 credits	

V	00	00	04	--	--	00	04 –07 courses 16-28 credits	86 credits
VI	00	00	04	--	-	00	04 –07 courses 16-28 credits	
VII	00	00	00	--	--	06	00 – 05 courses 06-26 credits	
VIII	00	00	00	--	--	08	00 – 05 courses 08-28 credits	
TOTAL CREDITS								170

- ED : At least 4 courses (16 credits)
- EO- Sciences : At least 1 courses (04 credits)
- EO-SS & Mgmt : At least 2 courses (08 credits)
-

2.3 COURSE CODE NOMENCLATURE

The courses of various B.Tech programmes shall be assigned a course code as per the following nomenclature

2.3.1 COURSE/DEPARTMENT/SPECIALIZATION/BRANCH CODING .

The courses of various B.Tech programmes shall be assigned a course code as per the defined nomenclature (Given later). This nomenclature shall use course/department/specialization/branch coding which are defined as given below.

TABLE 4: COURSE/DEPARTMENT CODES

XX	Course Category Code	FC	Foundation Core
		FE	Foundation Elective
		EO	Open Elective
YY	Name of Department Code	BT	Bio Technology
		CH	Chemistry
		CP	Computer Engineering, East Campus
		CS	Computer Science & Engineering
		CW	Civil Engg, West Campus
		EE	Electrical Engineering
		EC	Electronics & Communication Engineering
		EP	Electronics & Communication Engineering, East Campus
		HS	Humanities
		IC	Instrumentation & Control Engineering
		IT	Information Technology
		IW	Information Technology, West Campus
		ME	Mechanical Engineering
		MG	Management
		MP	Manufacturing Process & Automation
		MT	Mathematics
		MW	Mechanical Engineering, West Campus
		PD*	Personality Development
		PH	Physics

Note : Second Alphabet P indicates East Campus, and W indicates West Campus

TABLE 5 : B.TECH SPECIALIZATION/BRANCH CODES

ZZ	BT	Bio Technology
	CA	Computer Science & Engineering with Artificial Intelligence
	CB	Computer Science and Engineering (Big Data Analytics) (CSDA)(NSUT EAST CAMPUS)
	CD	Computer Science and Engi)neering (Data Science) (CSDS)
	CE	Civil Engineering (CE)(NSUT WEST CAMPUS)
	CG	Geoinformatics (GI)(NSUT WEST CAMPUS)

	CI	Computer Science and Engineering (IOT) (CIOT)(EAST CAMPUS)
	CM	Maths & Computing
	CO	Computer Science & Engineering, Main Campus
	EA	Electronics and Communication Engineering (Artificial Intelligence and Machine Learning) (ECAM) (NSUT EAST CAMPUS)
	EC	Electronics & Communication Engineering
	EI	Electronics & Communication Engineering (Internet of Things)
	EE	Electrical Engineering
	IC	Instrumentation & Control Engineering
	II	Information Technology (Internet of Things) (IIOT)(NSUT WEST CAMPUS)
	IN	Information Technology (NetwoRk security)
	IT	Information Technology
	ME	Mechanical Engineering
	MP	Manufacturing Process & Automation
	MV	Mechanical Engineering (Electric Vehicles) (MEEV)(NSUT WEST CAMPUS)

2.3.2 B.TECH COURSE CODE NOMENCLATURE

FOUNDATION CORE AND ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department Code			Course No.	
X	X	Y	Y	0	*	*

** can take numeric values only

XX and YY maybe chosen as given in Tables 1,2:

*PD offers FE courses like music, dance, yoga, sports, NSS, etc. A BOS for FE courses of PD nature (like Music, Dance, Yoga, NSS, etc), has be constituted with Dean Academics as the chairperson.

OTHER CORE AND ELECTIVE COURSES:

Program Code		Offering Department Code		Course Category	Course No.	
Z	Z	Y	Y	C/E	*	*

** can take numeric only;

C for Core and E for Elective (Discipline Centric);
YY and ZZ maybe chosen as given in Tables 1,2.

2.3.3 MOOCS (NPTEL BASED) FOUNDATION ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department (NPTEL) Code		UG/PG	Course No.	
X	X	F	F	G	*	*

** can take numeric only;

XX	Course Category Code	FE	Foundation Elective
		EO	Open Elective
FF		NH	Humanities & Social Sciences

	Name of Offering Department Code for NPTEL	NM	Management
		NP	Personality Development
		NS	Sciences
G	UG/PG	0	B.Tech
		1	M.Tech

2.3.4 STUDENT ROLL NUMBER NOMENCLATURE:

Students shall be assigned roll numbers as given below.

1. B.Tech:

Year of Admission	U	ZZ (FROM TABLE 2)	4 DIGIT NUMBER

3 SEMESTER WISE COURSE ALLOCATION

3.1 COURSE ALLOCATION FOR SEMESTER I

Refer to the separate manual on Equivalent courses and scheme of courses with pre-requites if you want to find the linked courses and pre_requisites for opting for EG courses.

B.Tech -SEMESTER I														
Course Code	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
FCMT001	FC	Mathematics-I	3	1	0	4	25	25	50	-	-	Maths	BASIC SCIENCES
FCCS002/FCHS005	FC	Computer Programming /English	3	0	2	4	15	15	40	15	15	COE/ IT	ENGG SCIENCES/ HUSS
FCEC003	FC	Electronics and Electrical Engineering	3	0	2	4	15	15	40	15	15	ECE/ ICE /EE	ENGG SCIENCES
FCPH004/ FCCH008	FC	Physics/Environment Science and Green Chemistry	3	0	2	4	15	15	40	15	15	PHYSICS / CHEMISTRY	BASIC SCIENCES
FCME006	FC	Basics of Mechanical Engg.	4	0	0	4	25	25	50	-	-	MPAE/ME	ENGG SCIENCES
FEXXxxx 2*	FE	Elective Foundation	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			28 contact hours 2*			20							
1: One week induction program as per AICTE norms. Classes for I semester will commence one week later.													

2*: The course codes, LTP distribution and Evaluation Scheme for Foundation Electives are given in Table 3 (list under preparation). The actual weekly load depends upon the Foundation Elective Course.

Students of the Departments of Group I shall be offered courses as follows:

1. Semeter I : Computer Programming, Physics
2. Semeter II : English, Environment Science and Green Chemistry

Students of the Departments of Group II shall be offered courses as follows:

1. Semeter I : English, Environment Science and Green Chemistry
2. Semeter II : Computer Programming, Physics

3.2 COURSE ALLOCATION FOR SEMESTER II

B.Tech. ECE SEMESTER II													
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory		Practical				
							CA	MS	ES	CA	ES		
FCHS005/FCCS002	FC	English/Computer Programming	3	0	2	4	15	15	40	15	15	COE/IT	ENGG SCIENCES/ HUSS
FCMT007	FC	Mathematics-II	3	1	0	4	25	25	50	-	-	MATHS	BASIC SCIENCES
FCCH008 / FCPH004	FC	Environment Science and Green Chem./ Physics	3	0	2	4	15	15	40	15	15	CHEMISTRY/ PHYSICS	BASIC SCIENCES
ECECC01	CC	Active Circuit Analysis and Synthesis	3	1	0	4	25	25	50	Nil	Nil	ECE	PROGRAM CORE/ ENGG SCIENCES
ECECC02	CC	Electronic Devices and Circuits	3	0	2	4	15	15	40	15	15	ECE	
ECITC03	CC	Data Structures and Algorithms	3	0	2	4	15	15	40	15	15	IT	
			24 2*			24							
2*: The actual weekly load depends upon the Core Courses offered by the Department													

3.2 COURSE ALLOCATION FOR SEMESTER III

B.Tech. SEMESTER III													
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
ECMTC04	CC	Mathematics for Communication & Signal Processing	3	1	0	4	25	25	50	-	-	MATHS	BASIC SCIENCES
ECECC05	CC	Signal and Systems	3	1	0	4	25	25	50	-	-	ECE	PROGRAM CORE
ECECC06	CC	Probability Theory and Random Process	3	1	0	4	25	25	50	-	-	ECE	PROGRAM CORE
ECECC07	CC	Microelectronics	3	0	2	4	15	15	40	15	15	ECE	PROGRAM CORE
ECECC08	CC	Digital Circuits and Systems	3	0	2	4	15	15	40	15	15	ECE	PROGRAM CORE
FE***02*	FE	Foundation Elective	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			20*			20							
*: The actual weekly load depends upon the Core Courses defined by the Department													

3.3 COURSE ALLOCATION FOR SEMESTER IV

B.Tech. SEMESTER IV													
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
ECCOC09	CC	Machine Learning and AI	3	0	2	4	15	15	40	15	15	COE	ENGG SCIENCES
ECECC10	CC	Electronic Design W/S	2	0	4	4	30	-	20	30	20	ECE	PROGRAM CORE
ECECC11	CC	Communication Engineering	3	0	2	4	15	15	40	15	15	ECE	PROGRAM CORE
ECECC12	CC	Micro-Processors and Comp. Architecture	3	0	2	4	15	15	40	15	15	ECE	PROGRAM CORE
ECECC13	CC	Electromagnetics	3	0	2	4	15	15	40	15	15	ECE	PROGRAM CORE
FE***03#	FE	Foundation Elective	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			20#			20							
#: The actual weekly load depends upon the Core Courses defined by the Department													

4. SYLLABUS OF COURSES

4.1 SYLLABUS OF FOUNDATION COMPULSORY COURSES

Course No.	Title of the Course	Course Structure	Pre-requisite
FCMT001	Mathematics - I	3L - 1T - 0P	None

COURSE OUTCOMES (COs):

1. Analyze and test Infinite Series and its convergence,
2. Successive differentiation and expansion of the function,
3. Curvature and Radius of Curvature in different coordinate systems,
4. Applications of definite integral,
5. Consistency of system of equations, Eigenvalue and Eigenvector.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Infinite Series: Tests for convergence of series: p-series (with proof), Comparison of ratios, Ratio, Integral, Raabe's, Logarithmic and Cauchy's nth root (all tests without proofs), Alternating series, Absolute convergence, Conditional convergence. Function of Single

UNIT-II

Variable: Hyperbolic functions, inverse hyperbolic function, successive differentiation, Leibniz theorem, Taylor's and Maclaurin's theorems (without remainder terms).

UNIT-III

Curvature: Polar Curves, Differential coefficients of length of arc, Cartesian, polar and parametric forms, pedal equation, Angle between tangent and radius vector, Curvature and Radius of Curvature in Cartesian, polar and pedal forms.

UNIT-IV

Applications of definite integral: Asymptotes (in Cartesian), elementary knowledge of curve tracing, area, length, surface area and volume of revolution (in Cartesian, parametric and polar co-ordinates).

UNIT-V

Matrices: Elementary row transformation, Rank of matrix, consistency and inconsistency of system of simultaneous equations, solution of non-homogeneous and homogeneous equations, Eigenvalue and Eigenvector, Characteristic equation, Cayley-Hamilton theorem. Modal matrix

SUGGESTED READINGS:

1. Calculus and Analytic Geometry by G.B. Thomas (Pearson Education)
2. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Publication)
3. Advanced Engineering Mathematics by Michael Greenberg (Pearson Education)
4. Advanced Engineering Mathematics by R. K. Jain and S.R.K. Iyenger (Narosa Publication)
5. Higher Engineering Mathematics by B. S. Grewal (Khanna Publication)

Course No.	Title of the Course	Course Structure	Pre-requisite
FCCS002	Computer Programming	3L - 0T - 2P	None

COURSE OUTCOMES (COs):

1. To understand the basic terminology and program structures used in computer programming to solve real world problems.
2. To understand the need for continuing to learn new languages to solve complex problems in different domains.
3. To learn the process of representing problems and writing, compiling and debugging programs.
4. To develop programming skills in using different types of data, decision structures, loops functions, pointers, data files and dynamic memory allocation/de-allocation.
5. To be able to code using Procedural and Object-Oriented languages.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Basics of C: Basic features of C Language like Identifier, Keywords, Variable, data types, Operators and Expression, basic screen and keyboard I/O, Control Statements, iteration, nested loops, Enumerated data types, bitwise operators, C Preprocessor statements. [6 hours]

UNIT-II

Arrays and Pointers: One and multidimensional dimensional arrays, strings arrays, operations on strings, Array and Pointers, Pointer to Pointer, other aspect of pointers, User Defined Data Types: Structures, Unions. [6 hours]

UNIT-III

Functions: Concept of modular programming, Using functions, Scope of data, Recursive functions, Pointers and functions, Command line arguments.

Files: Types of files, working with files, usage of file management functions.

[6 hours]

UNIT-IV

Overview of Object Oriented Programming: Python Programming, Concepts and Terminology. Data Types and Collection Data Types: Identifiers and keyword, Integral types floating point types, operations and formatting, Sequence types, Tuples, named Tuples, lists, set Types, sets, frozen sets, mapping types, Dictionaries, Iterating and Copying collections iterators and interactable operations and functions copying collection.

Central Structures and Functions: Conditional branching, looping, Exception handling catching and raising exceptions, custom exceptions custom functions, Names and Docstrings, Argument and Parameter unpacking, Accessing variables in Global scope, lambda functions. [9 hours]

UNIT-V

Modules and Packages: Packages, custom modules, overview of python's standard library, string handling, mathematics and Numbers, Times and dates, File formats, Data persistence.

File Handling: Writing and Reading binary data, raw binary data, compression, parsing text files, Random Access binary files, generic binary record file class.

[9 hours]

Guidelines for practical work:

Programs based on concepts of above languages.

SUGGESTED READINGS:

1. B. W. Kernighan and D.M. Ritchie, "The C programming language", Prentice Hall.
2. Herbert Schildt and Tata McGraw Hill, "The Complete Reference".
3. O Reilly Learning Python
4. Programming in Python 3: A Complete Introduction to the Python Language Pearson by Mark Summerfield

Course Type	Title of the Course	Credits	Course Structure	Pre-Requisite
FCEC003	ELECTRONICS AND ELECTRICAL ENGINEERING	4	3-0-2	None

Course Outcomes:

1. To understand the basics of AC and DC circuits, transformers along with DC generator and motor
2. To analyze series-parallel RLC circuits and
3. To implement basic circuits using diodes, BJTs and op-amps as circuit elements
4. To get familiarized with OP-AMP and its applications
5. To develop circuits using basic electrical and electronic components

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT

Unit-I

Electric Circuits: Basic Circuit Elements, Nodal and Loop Analysis,

Superposition, Thevenin's Theorem & Norton's Theorem and Maximum Power Transfer Theorem;

Unit-II

Steady-state analysis of AC circuits: Sinusoidal and phasor representation of Voltage and current, single phase AC circuit, behavior of R, L and C

Combination of R, L and C in series and parallel, Resonance; Introduction to three-phase circuits, Star-Delta Transformation

Unit-III

Transformers: Principle of operation and construction of single-phase transformer, Introduction to DC Motor.

Electronics Devices and Circuits: Junction Diode, Applications: rectifiers, clipping and clamping circuits, LEDs;

Unit-IV

Bipolar-junction Transistor: Physical operation, operating point, load-line, Self-bias circuit, single-stage CE amplifier configuration

Ideal op-amp, inverting, non-inverting and unity gain amplifiers, integrator, differentiator, summer/subtractor.

Unit-V

Digital circuits- Boolean Algebra, logic gates, K-Maps upto 4-variables, Combinational circuits: Adders and subtractors.

Flip-Flops: SR, JK, D, T and their characteristic tables. Introduction to Sensors, Introduction to Embedded Computers.

List of experiments for Electrical and Electronics Engineering

1. Verification of Maximum Power Transfer theorem
2. Verification of Thevenin's and Norton's theorems
3. Study of resonance in series RLC and parallel RLC circuits
4. Analysis of step-up and step-down transformer
5. Implement of series RC circuit as differentiator and integrator. Also perform their analysis as low pass and high pass filters
6. Implementation of clipping and clamping circuits
7. Implementation of half-wave and full wave rectifier circuits
8. Application of LEDs in electronic circuits
9. Implementation of CE amplifying configuration. Plot gain vs frequency graph
10. Implementation of Adders and subtractors.
11. Implementation of JK and Toggle flip-flops. Subsequently implement 3-bit asynchronous up-counter.
12. Measurement of power in single phase circuits using three voltmeter and three ammeter method.
13. Experiments with common sensors
14. Experiment with embedded computers

Suggested Reading:

1. M.E. Van Valkenburg, "Network Analysis" Pearson publishers, 3rd Edition
2. Boylestad and Nashelsky, "Electronic Devices and Circuit Theory" Pearson publishers, 10th Edition
3. Edward Hughes, "Electrical and Electronic technology", Pearson publishers, 10th Edition
4. Malvino and Leach, " Digital Principles and Applications", TMH publishers, 8th Edition

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCPH004	Physics	3L-0T-2P	None

COURSE OUTCOMESS (CO):

1. Knowing important concepts and phenomena linked to relativity
2. The concept of waves and oscillations are useful for doing analytical and numerical calculations for measurements, observations and gravitational wave communications.
3. The course is helpful to the students in understanding various optical wave phenomena which are required for optical & electromagnetic wave communications and in optical devices.
4. To develop the basic understanding of laser for gaining advance knowledge in the field of optical communication and opto-electronics.
5. The Concepts of Optical Fiber for modern developments in physics which are helpful in designing and developing new devices used in optical communications, medicine, environment, Industries and related physics.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	2	2	2	2	1	-	-	-	-	-	-
CO 2	3	2	2	2	2	1	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	3	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Relativity: Special Relativity, Lorentz Transformations, Velocity addition, Time dilation, Length Contraction, Variation of mass with velocity, Mass and energy, Relativistic momentum and relativistic energy, General theory of relativity, Einstein's theory of Gravitation, Gravitational waves, Gravity and Light.

UNIT-II

Oscillations and Waves: Damped and forced oscillations, Sharpness of resonance, Q-factor, Application in resonance, Acoustic waves, Pressure wave equations, Intensity pressure relation, Acoustic impedance, Reflection and transmission of acoustic waves, Impedance matching; Ultrasonics and its applications.

UNIT-III

Optics: Interference: Interference due to thin films, Newton's rings, and determination of the wavelength of sodium light, Interference due to wedge shaped film. Diffraction: Fraunhofer diffraction due to single slit and N Slits, Plane transmission grating, Rayleigh criterion of resolution, Resolving power of a grating, Polarization: Polarization in light, Birefringence, Nicol prism, Quarter and half wave plates, Production and analysis of plane, Circularly and elliptically polarized light, Optical rotation, specific rotation, Polarimeter.

UNIT-IV

Lasers: Absorption and emission of radiation, Main features of a laser, Spatial and temporal coherence, Einstein Coefficients, condition for light amplification, Basic requirement for Laser, Population Inversion - Threshold Condition, Line shape function, Optical Resonators, Three level and four level systems. Classification of Lasers: Solid State Laser-Ruby laser and Gas Laser- He-Ne laser (Principle, Construction and working), Optical properties of semiconductor, Semiconductor laser (Principle, Construction and working), Applications of lasers in the field of medicine, Industry, Environment and Communication.

UNIT-V

Fibre Optics: Need for fiber Optic Communication, Physical nature of Optical fiber, Theory of Light propagation in optical fiber, Acceptance angle and numerical aperture, Step index and graded index fibers, Single mode and multimode fibers, Losses in optical fiber, Optical Fiber cables and bundles, Dispersion in optical fibers: Intermodal and Intramodal dispersion.

List of Experiments:

1. To determine the value of "g" by Bar Pendulum and find the radius of gyratic
2. To determine the wavelength of He-Ne laser.
3. To find the numerical aperture and angle of acceptance of optical fiber.
4. To find the resolving power of a telescope.
5. To find the wavelength of sodium light by Newton's ring.
6. To find the wavelength of sodium light by Biprism.
7. To find the wavelength of Mercury green line by diffraction grating us spectrometer.
8. To find the focal length of combination of two lenses by Nodal slide assembly verify the formula.
9. To find the specific rotation of canesugar by polarimrter.
10. To find the dispersive power of prism material using spectrometer.

Text Books:

T1 Arthur Beiser, Shobhit Mahajan, `` Concepts of Modern Physics," Mc-Graw Hill
T2 D S Mathur, ``Mechanics," S Chand & co.
T3 N. Subramaniam and Brij Lal, ``A Text Book of Optics," S Chand &Co.
T4 A K Jha "A Text Book of Applied Physics, Volume-1" I.K. International Publishing House.
T5 Indu Prakash, ``A Text Book of Practical Physics, Volume-1," Kitab Mahal Publication.

Reference:

R1 Serwey, Moses, Moyer, ``Modern Physics," Cengage Learning
R2 Jenkins and White, ``Fundamentals of Optics," McGraw Hill
R3 Ajay Ghatak "Optics" McGraw Hill

REVISED SYLLABUS OF CORE ENGLISH

Course No.	Title of the Course	Course Structure	Prerequisite
FCHS 005	Core English	3L 0T 2P	None

Course Outcomes

CO 1: Acquire competence in Basic English grammar. Grammatical accuracy, avoiding inappropriacy and using language naturally and confidently

CO 2: Improve in the four integral skills of language and to be able to use language as a tool for effective communication

CO 3: Enable the learner to express and be understood by others with clarity and precision, in both written and spoken forms, minimizing ambiguity and verbosity.

CO 4: Understand creative use of language through translation, articles and paragraph writing.

CO 5: Reading: Encouraging the habit of reading for different purposes and to analyse, paraphrase and read critically.

CO 6: Develop competence in formal Standard English pronunciation and usage

CO 7: Build confidence to use a standard spoken form of English to face job interviews, and workplace interactive situations besides enabling the learner to pursue advanced professional courses.

COs.	Theory	Hours	Lab
1.	1. Vocabulary Enhancement CO 1 1.1 Using a standard dictionary- word spellings, meanings, usage, pronunciation, making sentences 1.2 Word collocations 1.3 Commonly misused words, verbal reasoning 1.4 One word substitutions 1.5 Abbreviations & foreign phrases	4	Lab Activity No 1: Phonetics: Sounds Used in English Language CO 6 Lab Activity No 2: Reading from newspapers/magazines/blogs to build up a repertoire of words CO 5

2.	2. Remedial & Applied Grammar CO1 & CO 2 2.1 Tenses & Voice 2.2 Subject-Verb Agreement 2.3Narration,Interrogative structures and Question tags 2.4 Prepositions, Pronouns and Adverbs 2.5 Redundancy 2.6 Idiomatic use of language 2.7 Identification of errors and editing	6	Lab Activity No 3: Introducing Oneself: Breaking the Ice CO 5 Lab Activity No 4: Situational & Spontaneous English(tense, registers) through Role Play CO 7 Lab Activity No 5: Question Formation & Mock Press Conference CO 5
3.	3 Techniques of Good Writing CO 5 & CO 2 & 3 3.1Writing self assessment tasks 3.2 Precis writing and note-making. 3.3 Paragraph and Essay writing. 3.4 Article writing and summarizing	10	Lab Activity No 6: Blog Writing/Creating a Newsletter Lab Activity No 7: Script writing & enacting for a street play CO 6
4.	4. Business Communication: CO 4 & CO 3 4.1 Formal and Informal Letter writing 4.2 Statement of Purpose 4.3 Job application & CV (summary statement of academic & professional profiles) 4.4Power point presentations through relevant slides.	10	Lab Activity No 7: Communication at Workplace. Develop negotiating skills by using appropriate language of courtesy Lab Activity No 8: Recording individual efforts and holding paired interactions and Group Discussions Lab Activity No 9: Preparing and practising for Interviews. CO 7
5.	5.Written Comprehension CO 3 & 4 5.1 The ability to write after listening to and reading select speeches, news bulletins, presentations and answering questions based on what has been heard.	10	Lab Activity No 10:)Introduction to Podcast and Understanding Audio texts Lab Activity No 11: Declamation and/or speeches

	5.2 Reading the given texts to skim, scan, infer and answer comprehension questions. 5.3 Reading texts like case studies and project reports for critical assessment. 5.4 Book Review		Lab Activity No 12: Reading, GD and presentation based on listed texts CO 7

Course No	Title of the Course	Course Structure	Pre-Requisite
FCME006	Basics of Mechanical Engineering	L-T-P: 4-0-0	None

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To know force, its nature and applications.
2. To know the basic principles of civil and mechanical structures.
3. To understand the fundamentals of thermodynamics and fluid mechanics.
4. To know the working principles of IC Engines.
5. To understand the importance of different engineering materials.
6. To understand the different manufacturing processes and machining operations.
7. To know the use of Automation in manufacturing.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	1	1	-	-	-	-	-	-	-
CO 2	3	2	2	1	1	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	-	-	-	-	-	-
CO 6	3	2	2	2	2	1	-	-	-	-	-	-
CO 7	3	2	2	2	2	1	-	-	-	-	-	-

COURSE CONTENT

Group A

Unit-I

Introduction to Engineering Mechanics: Rigid and Elastic bodies, Force and its type, Law of parallelogram of forces, Triangle law of forces, Polygon law of forces, Lami's theorem, Laws of motion, Moment, Couple, Varignon's theorem, Conditions of equilibrium, Concept of free body diagram, Coulomb's friction, Plane trusses,

Analysis of trusses, Numerical problems. (6 Hours)

Unit-II

Introduction to Strength of Materials: Simple stresses and strains, Direct, shear, and volumetric stresses and strains, Hooke's law, Tension test, Elastic constants, Poisson's ratio, Factor of safety, Introduction to beam, Types of beams, Types of loads, Shear force and bending moment diagrams (SFD and BMD) for Simple and Cantilever beams under various loading conditions, Numerical problems. (6 Hours)

Unit-III

Introduction to Manufacturing Engineering: Classification and use of engineering materials, Basic principles and applications of methods of manufacturing such as casting, forming and joining; Working principles and applications of machining operations such as Turning, Thread cutting, Milling, Shaping, Grinding, etc., Use of automation in manufacturing. (6 Hours)

Group B

Unit-IV

Introduction to Thermodynamics: Thermodynamic system, Cycle, Path, Thermodynamic properties, Extensive and intensive properties, Thermodynamic equilibrium, Reversible and irreversible processes, isochoric, Isothermal, Isobaric, Isentropic and Polytropic processes, First law of thermodynamics applied to a cycle and process, Kelvin-Planck and Clausius statements of Second law of thermodynamics, Carnot cycle, Entropy, Clausius inequality, Internal combustion (IC) engines, IC engines terminology, Spark ignition (SI) and Compression ignition (CI) engines, Two and four stroke engines, Air standard cycles such as Otto, Diesel, Dual and Brayton cycles, Numerical problems. (12 Hours)

Unit-V

Introduction to Fluid Mechanics: Properties of a fluid, Density, Specific volume, Specific weight, Specific gravity, Kinetic and Kinematic viscosity, Pascal's law and its applications, Laminar and turbulent flow, Use of continuity equation and Bernoulli's equation, Numerical problems. (6 Hours)

SUGGESTED READINGS

1. Engineering Mechanics- Beer and Johnston, Pearson
2. Strength of Materials- D.K. Singh, CRC Press
3. Engineering Thermodynamics- Nag, McGraw-Hill
4. Fluid Mechanics- Cengel, McGraw-Hill
5. Fundamentals of Manufacturing Engineering- D.K. Singh, CRC Press

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCMT007	Mathematics II	3L-1T-0P	None

COURSE OUTCOMES (CO)

1. Ordinary Differential Equations,
2. Partial Derivatives, Maxima and Minima for functions of two or more variables,
3. Evaluation of double and triple integral,
4. Concept of Numerical Methods and its Applications,
5. Concept of Probability and Statistics and its Applications.

COURSE CONTENT:

UNIT-I Ordinary Differential Equations:
Second & higher order linear differential equation with constant coefficients, general solution of homogenous and non-homogenous equations, Euler-Cauchy equation, Series solution by Frobenius method.

UNIT-II Function of Several Variables:
Partial Derivatives, Euler's Theorem, Total differentiations, Change of Variables, Jacobian and its basic properties, Taylor's theorem, Maxima and Minima for functions of two or more variables, Lagrange's method of undetermined multipliers.

UNIT-III Multiple Integrals:

Evaluation of double integral (in Cartesian and polar co-ordinates), change of order of integration, change of variables. Triple integral (in Cartesian) and its applications. Gamma and beta function.

UNIT-IV Numerical Methods:
Solution of system of linear equations using Gauss elimination method, LU decomposition method Gauss Seidel iteration method, Solution of polynomial and Transcendental equations by Newton-Raphson method, Numerical Integration by trapezoidal rule and Simpson's 1/3 and 3/8 rule, Numerical Solutions of first order ordinary differential equations: Euler's method, Runge-Kutta method of fourth order.

UNIT-V Probability and Statistics:
Conditional probability, Random Variables, Probability distribution functions-binomial, Poisson, exponential, uniform and normal distributions; Correlation, rank correlation and regression analysis; Sampling Theorem.

Recommended Books:

1. Calculus and Analytic Geometry by G.B. Thomas (Pearson Education)
2. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Publication)
3. Advanced Engineering Mathematics by Michael Greenberg (Pearson Education)
4. Advanced Engineering Mathematics by R. K. Jain and S.R.K. Iyenger (Narosa Publication)
5. Higher Engineering Mathematics by B. S. Grewal (Khanna Publication)
6. Probability and Statistics for Engineers by Anthony J. Hayter (Cengage Learning)
7. Numerical Methods for Scientific and Engg. Computations by M. K. Jain, S. R. K. Iyenger and R. K. Jain, (Wiley Eastern Ltd.)

4.2 FOUNDATION ELECTIVE COURSES

4.2.1 LIST OF FOUNDATION ELECTIVES

Table 3: FOUNDATION ELECTIVES

Code	Name of Foundation Elective	L T P Allocation			Evaluation Scheme Theory Practical					Pre-Requisites
		L	T	P	CA	MS	ES	CA	ES	
FEPD001	Sports-I	0	0	4	-	-	-	100	-	None
FEPD002	Sports-II	0	0	4	-	-	-	100	-	FE001
FEPD003	NSS	0	0	4	-	-	-	100	-	None
FEPD004	NCC	0	0	4	-	-	-	100	-	None
FEMG005	Corporate Social Responsibility	2	0	0	100	-	-	-	-	None
FEPD006	Music	0	0	4	-	-	-	100	-	None
FEHS007	Basic of social sciences	2	0	0	100	-	-	-	-	None
FEHS008	Spoken Skills in English	0	0	4	-	-	-	100	-	None
FEMG009	Financial Literacy	2	0	0	100	-	-	-	-	None
FEHS010	Introduction to Ethics	2	0	0	100	-	-	-	-	None
FEHS011	Stress Management	1	0	2	50	-	-	50	-	None
FEHS012	Organizational Behavior	2	0	0	100	-	-	-	-	None
FEPD013	Theatre	0	0	4	-	-	-	100	-	None
FEPD014	Dance	0	0	4	-	-	-	100	-	None
FEPD015	Yoga	0	0	4	-	-	-	100	-	None
FEPD016	Digital Film Making	0	0	4	-	-	-	100	-	None
FEPD017	Workshop (Electrical and Mechanical)	0	0	4	-	-	-	100	-	None
FEHS018	Ethical Decision Making	2	0	0	100	-	-	-	-	None

4.2.2 SYLLABI OF FOUNDATION ELECTIVES

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD001	Sports-I	0L-0T-4P	None

COURSE OUTCOMES (CO):
To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.

COURSE CONTENT: (Any 2 out Of 5 Components)

A. INTRODUCTION TO PHYSICAL EDUCATION IN THE CONTEMPORARY CONTEXT (Any Two)
1. Learn and demonstrate the technique of Suryanamaskar.
2. Develop Physical Fitness through Calisthenics / Aerobics / Circuit-Training / Weight-Training and demonstrate the chosen activity.
3. Select any one game available in the college and learn different techniques involved in its play

B. CORE PHYSICAL EDUCATION-: FITNESS, WELLNESS AND NUTRITION (Any Two)
1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups Muscular Endurance); Harvard Step Test, Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measuring height, weight, waist circumference and hip circumference, Calculation of BMI (Body Mass Index) and Waist-Hip Ratio
3. Engage in at least one wellness programme and write a report on it.

C. CORE PHYSICAL EDUCATION-: POSTURE, ATHLETIC CARE AND FIRST AID (Any Two)
1. Demonstrate Stretching and Strengthening Exercises for Kyphosis, Scoliosis, Lordosis, Knock Knees, Bow Legs, Flat Foot, Back Pain and Neck Pain
2. Illustration and Demonstration of Active and Passive Exercises
3. Asanas with Therapeutic Value (Any five asanas): Karnapeedasana, Padmasana, Dhanurasana, Sarvangasana, Paschimottanasana, Chakrasana, Halasana, Matsyasana, Ardhmatsyendrasana, Usthrasana, Mayurasana, Shirshasana, Vajrasana.
4. Practice P.R.I.C.E. in First Aid.

D. SPORTS ADMINISTRATION & MANAGEMENT (Any Two)
1. Demonstration of Supervision activities in Sports Management.
2. Demonstration of skills of Management.
3. Demonstration of fixtures of various kinds in sports competitions.
4. Demonstration of technical and non-technical purchase procedure.
E. Adventure Sports and Leadership Training

SUGGESTED READINGS:
1. Graham, G., ``Teaching Children Physical Education: Becoming a Master Teacher. Human Kinetics,'' Champaign, Illinois, USA.
2. Corbin, C. B., G. J. Welk, W. R Corbin, K. A. Welk, ``Concepts of Physical Fitness: Active Lifestyle for Wellness,'' McGraw Hill, New York, USA.
3. Anspaugh, D.J., G. Ezell and K.N. Goodman, `` Teaching Today Health,'' Mosby Publishers
4. Beotra, Alka, ``Drug Education Handbook on Drug Abuse in Sports,'' Applied Nutrition Sciences, Mumbai.
5. Ammon,R., Southall , R.M. and Blair, D.A., ``Sports Facility Management, ''West Virginia, USA: Fitness Information Technology Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD002	Sports-II	0L-0T-4P	FE001

COURSE OUTCOMES (CO):
To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.

COURSE CONTENT:
(Any 3 out Of 5 Components)
A. Sports for all (Any Two)
1. To participate in any intramural Tournaments (one team game and one Individual Game) of choice.
2. To participate/ attend at least 15 hours in Fitness training at Field or at Gymnasium.
3. Participate in at least one track and one field event on Annual Sports day.
4. To participate in Inter College Tournament

B. Skill Enhancement Courses (any one out of three)
1. Wellness and Fitness
2. Holistic personality Development
3. Sports Journalism
4. Mass demonstration Activities

C. MEDIA AND CAREERS IN PHYSICAL EDUCATION (Any Two)
1. Organize an event / intramural / tournament in your college.
2. Prepare a News Report of an observed Sports competition.
3. Create a presentation on any topic from Physical Education using an audio-visual aid.
4. Demonstrate Warming-up / Conditioning / Cooling-down exercises.

D. MANAGEMENT OF AEROBICS & GROUP TRAINING (Any Two)
1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups (Muscular Endurance); Harvard Step Test or Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measurement of Pulse Rate / Heart Rate at Radial Artery and Carotid Artery, Calculation of Target Heart Rate
3. Developing a 5-10 minute routine of aerobics with appropriate music for each component of health related physical fitness

E. SPORTS INDUSTRY & MARKETING (Any Two)
1. Identify an issue or a trend in the sports industry: o Players in professional or college sports o Ownership
2. Marketing Plan: Environmental Factors and Product Plan Draft, Paper bibliography/works cited.
3. Sponsorship proposal
4. Developing a budget plan for an event
5. Athlete branding

SUGGESTED READINGS:
1. Covey, S. , `` 7 Habits of Highly Effective People, '' Covey Publications, USA
2. Magill, R.A., `` Motor Learning and Control: Concepts and Applications," McGraw Hill Publication.
3. Masteralexis, L.P., C. Barr and M. Humms, ``Principles and Practices of Sport Management," Jones and Bartlett Publisher
4. Bishop, J.G., ``Fitness through Aerobics," Benjamin Cummings USA.
5. Brown K.M., `` Physical Activity and Health: An Interactive Approach," Jones and Bartlett Publisher
6. Cornwell. T.B, `` Sponsorship in marketing: Effective communications through sports, arts and events, '' Routledge Publishers
7. DeGarris, L., ``Sports Marketing: A Practical Approach," Routledge Publishers, USA

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD003	**National Service Scheme (NSS)**	0L-0T-4P	None

COURSE OUTCOMES (CO):
1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;
4. Utilize their knowledge in finding practical solution to individual and community problems;
5. Develop capacity to meet emergencies and natural disasters

COURSE CONTENT:

Unit-I

Introduction to NSS: Orientation and structure of NSS, History of Social Reforms in Modern India: Brahmo Samaj, Arya Samaj, Satya Shodhak Samaj: Principles and Functions

Unit-II

Regular activities: Distribution of working hours- association between issues and programs- community project- urban rural activities, association- modes of activity evaluation

Unit-III

concept of society- development of Indian society: Features- Division of labors and cast system in India; Features of Indian constitution; Provisions related to social integrity and development

Unit – IV

N.S.S. Regular Activities

A) College campus activities
B) N.S.S.activities in Urban and Rural areas
C) Role of Non-Government Organisation (NGO) in social Reforms
i) Red Cross
ii) Rotary

SUGGESTED READINGS:

1. National Service Scheme Manual, Govt. of India
2. Training Programme on National Programme scheme, TISS.
3. Orientation Courses for N.S.S. programme officers, TISS.
4. Ram Ahuja, ``Social Problems in India,'' Rawat Publication.
5. History of Social Reforms in Maharashtra, Ed. J. Y. Bhosale, S. U. Kolhapur

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD004	**National Cadet Corps (NCC)**	0L-0T-4P	None

COURSE OUTCOMES (CO):
1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;

4. Utilize their knowledge in finding practical solution to individual and community problems;
5. Develop capacity to meet emergencies and natural disasters;

COURSE CONTENT:

UNIT I:
Introduction to NCC, National Integration & Awareness: Religions, Culture, Traditions and Customs of India, National Integration: Importance and Necessity, Freedom Struggle.

UNIT II:
Adventure Training: – Obstacle course, Slithering, Trekking, Cycling, Rock Climbing, Para Sailing, gliding, Scuba Diving- methods and use.

UNIT III:
Environment Awareness and Conservation: Natural Resources – Conservation and Management. Water Conservation and Rainwater Harvesting

UNIT IV:
Personality Development and Leadership: Introduction to Personality Development, Factors Influencing /Shaping Personality: Physical, Social, Physiological, Philosophical and Psychological, Self-Awareness Know yourself/ Insight, Change Your Mind Set, Communication Skills: Group Discussion / Lecturettes (Public Speaking), Leadership Traits, Types of Leadership

SUGGESTED READINGS:
1. Bhogle Anita & Bhogle Harsha, ``The Winning way, Learning from sports for managers,'' Westland Publications
2. Sharma Robin, `` The leader had no title, '' Simon and Schuster Ltd.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG005	Corporate social responsibilities	2L-0T-0P	None

COURSE OUTCOMES (CO):
1. The course will help students to understand corporate and emerging social responsibility for the corporate in reference to India and global situation
2. The course will support students to prepare themselves to work with corporate understanding collective aspiration of the society, individual and corporate social responsibility.

COURSE CONTENT:

UNIT I: Corporate social responsibility in Indian context and International: CSR – Definition, concepts, Approaches of CSR, overview of corporate social responsibility and corporate social accountability, SR Tools, National and International CSR

activities, corporate philanthropy, drivers of CSR, difference between corporate governance, corporate philanthropy and CSR

UNIT II: Business ethics and corporate social responsibility: Concept of business ethics – meaning, Importance and factors influencing business ethics. Corporate Governance – meaning, significance, principles and dimensions. Ethical decision – making in different culture, consumer protection, environment protection, gender issues in multiculturalism, ethics and corruption, ethics and safety. Business benefits of CSR

UNIT III: Legislative measures of CSR: Corporate, labor, stake holders, Environmental and pollution. Social Accounting, Social Auditing, SA: 8000 and Corporate Social Reporting.

SUGGESTED READINGS:

1. Harsh Srivastava,`` The business of social responsibility,'' books for change
2. CV. Baxi and Ajit Prasad,`` Corporate social responsibility – concepts and cases,'' Excel Books
3. Dr. M. Mahmoudi,`` Global strategic management,'' Deep & Deep Publications Pvt. Ltd.
4. S K. Bhatia,`` International Human resource management – Global perspective,'' Deep & Deep Publications Pvt. Ltd.
5. J.P. Sharma, ``Governance, Ethics and Social responsibility of business, '' Ane books Ltd.
6. Kotler Philip and Lee Nancy, `` Corporate social responsibility; doing the most good for your company,'' John Wiley
7. Simpson, Justine and Taylor, John R, `` Corporate Governance Ethics and and CSR,'' Kogan Page Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD006	**Music**	0L-0T-4P	None

COURSE OUTCOMES (CO):

The student will be familiarized with the basic terms used in Indian classical music. Also it familiarizes with the life history of some dignitaries in the field of music. This course also throws some light on the ancient music and its origins in India.

COURSE CONTENT:

Unit 1: Study of the following terms: - Mela (Thãt), ÃshrayRãga, Rãga, Lakshana, Shruti, Alankar, Gamak, Vadi-SamvãdiAnuvãdi-Vivãdi, VakraSwara, Varjit-Swara.

Unit 2: Biographies & contributions of the following: - Jaidev, MansinghTomar, Abdul Karim Khan, Tyagaraja, Pt. Bhatkhande, Pt. Ravi Shankar

Unit 3: Study of following Rãgas&TãlaRãga- Yaman, Jaunpuri, Khamaj. Tãla- Ektãl, Jhaptãl

Unit 4: Genara**l** discussion and definition of the following: -

a. Khyãl, MaseetKhani – Razakhani gat, Dhrupad, Tarana, Meend, Soot, Murki, Kan, Khatka, Krintan, Harmony, Melody.
b. Writing of Bhatkhande Swarlipi Paddhati.
c. Writing of Tãlasand Compositions in Notation.
d. Detailed study of Rãgas (Rãga- Bihag, Malkauns, Vrindavani Sarang) and comparative study of Rãgas.
e. Essay, Shastriya Sangeet (Classical Music) & SugamSangeet(Light Music)

Unit 5: Vedic Music – Samvedic Sangeet, Swara, Vadya, Bhakti, Vikãr. General study of Natyashastra, SangeetRatnakar.

SUGGESTED READINGS:

1. Vasant and Laxmi Narayan Garg,`` Sangeet Visharad," Sangeet Karyalay
2. Sarat Chandra Pranjpayee and Chowbhamda ,`` BhartiyaSangeetkaItihas," Surbharti Prakashan
3. Bharat Muni,`` NatyaShastra,"
4. Sharangdeva ,`` SangeetRatnakar,"
5. Sharad Chandra Pranjpayee ,`` Sangeet Bodh,"
6. Thakur Jaidev Singh ,`` Indian Music," Sangeet research academy
7. V. N. Bhatkhande,`` Mallika Part II & III," KramikPustak.
8. V. N. Patwardhan,`` RaagVigyan,"
9. RaginiTrivedi,`` Ragvibodha Mishrabani, Vol. I & II,"

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS007	**Basics of Social Sciences**	2L-0T-0P	None

COURSE OUTCOMES (CO):

Social science is a major category of academic disciplines, concerned with society and the relationships among individuals within a society. It in turn has many branches, each of which is considered a "social science".

COURSE CONTENT:

Unit I: Economics, political science, human geography, demography and sociology.

Unit II: Humanities, anthropology, archaeology, jurisprudence, psychology, history, and linguistic.

Unit III: Political science, economics, sociology, international politics and scientific methodology.

SUGGESTED READINGS:

1. A.C. Kapoor, "Principles of Political Science," S. Chand Publications
2. A.K. Sharma, "Issues in Social Demography," Mittal Publications
3. Kathy S. Stolley, "The Basics of Sociology," Greenwood Press.
4. Paul M. Muchinsky ,"Psychology Applied to Work," Thomson Learning Inc

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS008	**Spoken Skills in English**	0L-0T-4P	None

COURSE OUTCOMES (CO):

1. This course will focus on oral & presentation skills of students with practice sessions in the language lab.
2. This course will develop confidence building in oral skills of learners.
3. It will seek to encourage the day to day conversations/dialogues and communicative needs of learners with ample practice in the lab.
4. The theory class will boost practice in ample language exercises to encourage oral skills.
5. This will also involve practice sessions in interview skills, group discussions & pair work.
6. Basics of communication process, Barriers to Oral Communication
7. Elevator pitches - Practicals

COURSE CONTENT:

- Practice on listening and reading comprehension
- Language lab practice for group discussion and interviews
- Definition and discussion on communication & the barriers in communication with practical training to use language as a tool for sharing, discussing, handling and convincing others.

SUGGESTED READINGS:

Everyday English I & II Cambridge University Press/Foundation books

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG009	**Financial Literacy**	2L-0T-0P	None

COURSE OUTCOMES (CO):

1. To provide in-depth knowledge of the banking and Principles of Investment, financial planning.
2. Help students in understanding stocks, sell strategy, mutual fund options, investing in education, planning for the future, purchasing your first home, taxes and tax planning, life insurance options, health insurance, property insurance, estate planning, and keeping money in perspective.

COURSE CONTENT:

UNIT I: Banking- Definition, Role of Bank in growth of saving and Investment, Types of banks , Services offered by banks, Deposits and Loans, Types of A/c, Opening a bank A/c, How to Transact with banks, KYC norms, (A/c opening form, Address Proof), How to read bank statement, Banking products and services, Calculating Interests – Saving, FD, Simple and Compound Interest, Power of compounding Loans, Types of loans, taking a home loan, Definition of EMI, Calculation of EMI, Post office-Account and transactions, Basic of foreign Exchange, Importance and Use of Foreign Exchange, Regulator Role of RBI, mutual funds.

UNIT II: Investment: Principles of Investment – Safety, Liquidity and Return, Investment plans, Hybrid plans-Ulip, SIP and VIP of mutual funds, index funds

UNIT III: Financial Planning- Meaning, Household financial health checkup, Important life stages, Medical and other Emergencies, ; Insurance, Meaning, Need and Wants, Loss protection, Life, non-life and health, Benefits of Insurance, Term plans, Social obligations Budgeting, Buying a house, Plan a vacation, Retirement planning, Price of procrastination, Market and financial instruments, Primary market, Secondary market, Financial Statement analysis,

UNIT IV: Scams, Fraud Schemes-Insider trading, Money laundering; Consumer protection and redressal mechanism, Rights of Consumers, Applicable to financial services, Filing a complaint, Complain to entity concerned, Regulators, Arbitration, Consumer courts, Govt. Websites-(PG Portals), Investor Associations, Taxes, Meaning, Need of Taxes, Types of taxes, How taxes impact income, Income, wealth and gift tax, Service tax, STT, Stamp Duty, Tax planning v/s tax evasion, Tax rates, Tax free bonds, Tax saving investment

SUGGESTED READINGS:

1. Braunstein, Sandra, and Carolyn Welch, `` Financial literacy: An overview of practice, research, and policy," Fed. Res. Bull.
2. Cole, Shawn A., and Gauri Kartini Shastry,`` Smart money: The effect of education, cognitive ability, and financial literacy on financial market participation," Harvard Business School, 2009.
3. Study material of NSE.
4. Gitman, joehnk and Billingsley, ``Personal financial planning," Cengage Learning
5. Madura Jeff,`` Personal finance student edition," Prentice Hall PTR.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS010	Introduction to Ethics	2L-0T-0P	None

COURSE OBJECTIVES (CO):

1. It is aimed to comprehend right from wrong, to act upon something tricky with a deliberative analysis.

Course Outcomes:

2. Helps in addressing issues with a moral reasoning and analysis.

COURSE CONTENT:

1. Fundamentals of Ethics
2. Issues in Moral Philosophy
3. Theories of Justice and their Applications
4. Ethical Decision Making
5. Applied Ethics
6. Media Ethics
7. Environmental Ethics
8. Technology & Ethics
9. Feminism

SUGGESTED READINGS:

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS011	Stress Management	1L-0T-2P	None

COURSE OBJECTIVES (CO):

The objective of this course to help the students to understand the nature of stress, sources of stress and to identify the symptoms of stress through first unit. Second Unit aims to teach the students to learn certain skills and the strategies required for effectively managing the stress and ability to cope up from the stressful situations.

COURSE OUTCOMES (CO):

1. To understand the nature, sources of stress and consequences of stress
2. To overcome from the constraints in managing stress
3. To develop the motivation to adopt different technology

COURSE CONTENT:

UNIT I: Stress (GAS Model), Learning about sources of stress and its symptoms: Nature of stress- various sources of stress environmental, social (including social media), physiological and psychological; Symptoms of stress - emotional response, physiological & behavioral; relationship between stress and performance, relationship between stress and health

UNIT II: Learning to manage stress effectively: Methods - yoga, meditation, Vipassana,
relaxation techniques, clarifying problem, alternate actions, support (Problem focused) emotion focused constructive approach, Indian Case Studies

Practical: (50 marks)
Any two practical's based on Unit I and II

SUGGESTED READINGS:

1. DiMatteo, M.R. & Martin, L.R.(2002). Health psychology. New Delhi: Pearson. Neiten, W. & Lloyd, M.A (2007). Psychology applied to Modern life. Thomson Detmar Learning .
2. Taylor, S.E. (2006). Health psychology, 6th Edition. New Delhi: Tata McGraw Hill

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS012	**Organizational Behavior**	2L-0T-0P	None

COURSE OUTCOMES (CO):

1. The course aims at providing a comprehensive understanding of organization (structure and culture) and its functioning, at the levels of the individual, group and organization as a whole.
2. To acquaint the students with employee motivation and work attitudes and its relationship with performance and productivity.
3. To help students gain insights into the concept of organization change in the context of ever changing business environment and provide them with various tools of organizational development.

4. To provide students an overview of organizational dynamics in the light of power, politics and stress.

COURSE CONTENT:

Unit 1: Understanding Organizational Behavior: Defining organization and Organizational Behavior (OB), OB Model; the Organizational structure; Common Organizational Designs; New Design Options

Unit 2: Employee Attitudes and Motivation: Job Satisfaction; Organizational Commitment; Organizational Citizenship behavior; Positive Organizational Behavior (POB):
Theories of Work motivation: Content theory (Maslow, Herzberg), Process theory (Vroom's Expectancy Theory, Equity Theory), Contemporary theories (Goal Setting theory and Self-Regulation theory)

Unit 3: Dynamics of Organizational Behavior: Organizational culture; Power and Politics: Influence, sexual harassment, organizational politics, Causes & Consequences of stress at the workplace:

Unit 4: Organizational change and development: concept of organizational change, model of organizational change (one model), organizational development: concepts, models (one model), techniques of organizational development

Unit 5: Organizational behavior in startups

SUGGESTED READINGS:

1 Aamodt, M. G. (2016). *Industrial/Organizational psychology: An applied approach.* Boston: Cengage Learning.
2 Luthans, F. (2011). Organizational behaviour:An evidence based approach,12th Edition. McGraw Hill
3 Muchinsky, P.(2007). Psychology applied to work: An introduction to Industrial and Organizational Psychology. NC: Hypergraphic Press.
4 Pareek, U. &Khanna,S.(2012,Third edition). Understanding Organizational Behaviour. Oxford: Oxford University Press.
5 Pareek, U., & Gupta, R. K. (2010).*Organizational behaviour*. New Delhi: Tata McGraw Hill.
6 Robbins, S.P., Judge T.A. and Sanghi, S.(2009) Essentials of Organizational Behaviour,10th edition. Pearson Education , India.
7 Schultz, D & Schultz, S (2013).Psychology and Work Today. Pearson Education, India.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD013	**Theatre**	0L-0T-4P	None

COURSE OUTCOMES (CO):
Our goal is to nurture artist-scholars who are well read in dramatic literature, who understand the social and historical contexts of that literature, who appreciate contemporary performance and dance, who think critically, who master discipline-specific skills, and who make compelling artistic choices on stage.

COURSE CONTENT:

Unit 1: Concept of Acting in Indian Classical theatre. Western styles of theatre acting.

Unit 2: Basics of the following: Acting in Grotowski's Poor Theatre, Folk Theatre of India

Unit 3: Acting for Camera –Knowledge of camera frames and movement within the confines of a frame, blocking, difference between theatre and Camera acting, Concentration.

Unit 4: Acting consistently for different takes, acting scenes out of order, Auditions, acting exercises. Art of Dubbing.

SUGGESTED READINGS:

1. Boleslavsky, Richard,`` Acting: the First Six Lessons," New York Theatre Arts.
2. Hagen, Uta,`` Respect for Acting," Macmillan Press.
3. Hodge, Alison,`` Twentieth Century Actor Training," London and New York.
4. Routledge ,Stanislavski, Konstantin,`` An Actor's Work: A Student's Diary," Trans. and ed. Jean
5. Jeremiah Comey ,`` The Art of Film Acting," Focal Press .
6. Philips B Zarrilli,`` Acting (Re) Considered," Routeledge .
7. Cathy Hassey,`` Acting for Film," Allworth Press 9. Singh. Y,`` Indian Sociology social conditioning and emerging concerns," Vistaar publication.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD014	**Dance**	0L-0T-4P	None

COURSE OUTCOMES (CO):

This course will provide the student with the fundamentals necessary for advanced dance skills. Further, this course will develop student appreciation of dance as an art form and lifetime activity. Designed to familiarize students with technique, the student will also study vocabulary, different forms of dance, issues in dance and the history pertaining to the world of dance. The student will develop kinesthetic awareness, movement memory, creative abilities and aesthetic appreciation of various dance forms. The enhancement and the development and maintenance of physical fitness, self-confidence, self-discipline and independence with the body by providing informal showings during class are the goals expected to be achieved. Each student should leave this class having been encouraged, esteemed, and take with them a new appreciation of dance.

COURSE CONTENT:

- Basic workout
- Introduction to Hip Hop and B-Boying with a simple choreography
- Exercise like: Rolling, jumping, moving shoulders. Footwork, Floor steps, Beat knowledge.
- Freestyle combination along with House dance style.
- Expressions class: Body expressions, Face expressions.

- Introduction of Contemporary Dance. Basic exercise of Contemporary Dance. Exercise for flexibility, Floor steps, Spinning and Balancing.
- Introduction to Jazz. Basic exercise and proper routine practice.

SUGGESTED READINGS:

1. Jonathan Burrows, ``A Choreographer's Handbook,'' Routledge
2. Jacqueline M. Smith-Autard, ``Dance Composition: A Practical Guide to Creative Success in Dance Making,'' Routledge

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD015	**Yoga**	0L-0T-4P	None

COURSE OUTCOMES (CO):
Students will learn about the importance of yoga in their lives. They will be exposed various types of yoga, their health benefits.

COURSE CONTENT:

UNIT-I
Origin of Yoga & its brief development, Meaning of Yoga & its importance, Yoga as a Science of Art (Yoga Philosophy),Meaning of meditation and its types and principles.

UNIT- II
Classification of Yoga/Types of Yoga, Hatha Yoga , Raja Yoga, Laya Yoga, Bhakti Yoga, Gyan Yoga, Karma Yoga, Asthang Yoga.

UNIT –III
Principles of Yogic Practices, Meaning of Asana, its types and principles, Meaning of Pranayama, its types and principles, Meaning of Kriya its types and principles.

UNIT -IV
Yogic therapies and modern concept of Yoga, Naturopathy, Hydrotherapy, Electrotherapy, Messotherapy, Acupressure, acupuncture, Meaning and importance of prayer, Psychology of mantras, Different mudras during prayers

SUGGESTED READINGS:

1. William Broad,`` The Science of Yoga: The Risks and the Rewards,'' Simon and Schuster
2. Swami Vishnu Devananda,`` The Complete Illustrated Book of Yoga,'' Harmony

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD016	**Digital Film Making**	0L-0T-4P	None

COURSE OUTCOMES (CO):
Students will learn about various technicalities involved in digital film making. They will also expose to history of cinema, preproduction etc.

COURSE CONTENT:

Unit 1 – History of Cinema, Research & Script
Early Cinema, Development of Classical Indian & Hollywood Cinema, History of Global Film including European Film (1930-present), Origin of Classical narrative cinema-Soundless film, Exploration of film and analysis of the three-part beginning, middle and end of story, **Research**(Finding and Collecting materials and facts related to your story. Where and How to find the materials related to your story. Things to consider before sketching down your story), **Script (Scriptwriting Process and its various phases), Film Grammar for Scriptwriting.**

Unit 2 – Pre-Production
Digital Video Cinematography: Introduction to Digital Video Cinematography Cinematography, Interactivity and emotions through Cinematography,
Building blocks, Compositions, Lenses and Cameras, Types of lenses: Zoom Lens, Prime Lens, Types of Cameras: HD Cameras, Basics of Film Camera, Difference between, Film Camera and Digital Camera, DSLR and HDSLR Cameras, Lighting, Psychology of light, Visual Environment, Directional Effect of Light, Lighting design process, Three-point lighting, High-Key lighting, Low Key lighting, Construction of a Shot, Color, Contrast, Deep Focus, Shallow Focus, Depth of Filed, Exposure, Racking focus, Frame Rate, Telephoto shot, Zoom shot.

Unit 3- Digital Video Editing
Effective Editing, Principles of Video Editing, Non-Linear Editing (NLE) Concept, The Three-Point Edit, Non-Linear Editing (NLE) Techniques, Working in the Timeline, Transitions, Key framing, Applying Filters, Ingesting.

Unit-4Advanced Editing Techniques
NLE Compositing, Color Correction & Color Grading, Working on Audio, Titling

SUGGESTED READINGS:
1. Mark Brindle and Chris Jones, `` The Digital Filmmaking Handbook," Quercus

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD017	**Workshop (Electrical and Mechanical)**	2L-0T-0P	None

COURSE OUTCOMES (CO):
1. Student will be able to make various joints in the given object with the available work material.
2. The students will be able to understand various wiring connections

COURSE CONTENT:

Mechanical Workshop Experiments
1. Blacksmith
2. Carpentry
3. Fitting
4. Foundry
5. Welding

Electrical workshop Experiments

1. Study & Performance Of Different Types Of Wire Joints
2. Study And Performance Of Staircase Wiring
3. Study And Performance Of Series And Parallel Connection Of Flourescent Tube Light
4. Study And Performance Of Godown Wiring
5. Series And Parallel Connection Of Bulbs And Power Sockets By Single Switch And Multi Switches.

SUGGESTED READINGS:

1. Hajra Choudhury, Hazra Choudhary and Nirjhar Roy, ``Elements of Workshop Technology, vol. I, '' Media promoters and Publishers Pvt. Ltd.
2. W A J Chapman, Workshop Technology,`` Part -1, 1st South Asian Edition,'' Viva Book Pvt Ltd.
3. P.N. Rao, ``Manufacturing Technology, Vol.1,'' Tata McGraw Hill
4. Kaushish J.P., `` Manufacturing Processes, '' Prentice Hall

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS018	**Ethical Decision Making**	2L-0T-0P	None

COURSE OUTCOMES (CO):

COURSE CONTENT:

UNIT I: ETHICAL CONCEPTS AND ETHICAL APPROACHES

1. Values, Dilemma and Choices
2. Responsibility, Justice & Fairness
3. Respect for self and others

UNIT II: ETHICAL DECISION PROCESS

1. Ethical codes and tests
2. Steps to ethical decision-making
3. Case studies and Situational role plays

SUGGESTED READINGS:

- Blanchard, K., & Peale, N.V. (1988) The Power of Ethical Management, New York: William Morrow and Co. pp. 20-24. http://www.blanchardbowleslibrary.com/books/powerofethicalmanagement.htm
- Brown, M. (1996) The Quest for Moral Foundations: An Introduction to Ethics Georgetown University Press
- Davis, M. (1999) Ethics and The University, New York: Routledge.
- Heller, R. (1998) Making Decisions, New York: DK.
- Josephson, M. S. (2002) Making Ethical Decisions, Josephson Institute of Ethics.
- Kardasz, F. (2008) Ethics Training For Law Enforcement: Practices and Trends, VDM, Verlag Dr. M.ller.
- Nosich, G. M. (2002) Learning to Think Things Through: A Guide to Critical Thinking, Prentice Hall.

4.3 PROGRAM CORE COURSES

4.3.1 LIST OF PROGRAM CORE COURSES

SEM.	Code	Name of Core Course	L T P Allocation			Evaluation Scheme Theory Practical					Syllabus page Nos
			L	T	P	CA	MS	ES	CA(P)	ES(P)	
II	ECECC01	Active Circuit Analysis and Synthesis	3	1	0	25	25	50	Nil	Nil	40
	ECECC02	Electronic Devices and Circuits	3	0	2	15	15	40	15	15	
	ECITC03	Data Structures and Algorithms	3	0	2	15	15	40	15	15	
III	ECMTC04	Mathematics for Communication & Signal Processing	3	1	0	4	25	25	50	-	
	ECECC05	Signal and Systems	3	1	0	4	25	25	50	-	
	ECECC06	Probability Theory and Random Process	3	1	0	4	25	25	50	-	
	ECECC07	Microelectronics	3	0	2	4	15	15	40	15	
	ECECC08	Digital Circuits and Systems	3	0	2	4	15	15	40	15	
IV	ECCOC09	Machine Learning and AI	3	0	2	4	15	15	40	15	
	ECECC10	Electronic Design W/S	2	0	4	4	30	-	20	30	
	ECECC11	Communication Engineering	3	0	2	4	15	15	40	15	
	ECECC12	Micro-Processors and Comp. Architecture	3	0	2	4	15	15	40	15	
	ECECC13	Electromagnetics	3	0	2	4	15	15	40	15	

4.3.2 SYLLABI OF PROGRAM CORE COURSES : II SEMESTER

Course No.	Title of the Course	Credits	Course Structure	Pre-Requisite
ECECC01	Active Circuit Analysis and Synthesis	4	L-T-P 3-1-0	None

COURSE OUTCOMES (CO)

After completing the course, the students should be able to:

CO 1: Understand the concepts *pathological* elements and their applications in modelling various electronic devices and generation of equivalent circuits

CO 2: Be capable of analyzing passive and *active* circuits using Laplace transforms

CO 3: Determine various 2-port parameters of passive and *active* circuits and understand their applications

CO 4: Apply the basic concepts of graph theory in the formulation of linearly independent KCL and KVL equations

CO 5: Formulate state equations and determine their time domain solution

CO 6: Synthesize passive/*active* networks from the given driving-point/transfer functions

Unit Numbers	Topics
Unit1	*Active* elements: Gyrator, Impedance converters and inverters; Pathological elements: nullator, norator, nullor, current mirror, voltage mirror; representation of the four controlled sources by pathological models; nullor representation of ideal op-amp, ideal BJT and ideal MOSFET and their applications in realizing transistorized-gyrators and impedance converters/inverters; generation of equivalent realisations of *active* elements using pathological models
Unit 2	Circuit analysis using Laplace transform; impulse response and network function, unit impulse, unit step response and unit ramp responses; convolution integral and its application; transient analysis of passive and *active* networks; steady state and frequency domain analysis of passive and *active* circuits
Unit 2	Two port parameters: conversion of parameters, various types of inter-connections of two-port networks; Driving point and transfer functions of passive and *active* circuits, poles and zeroes, relation between location of poles and time response, stability of one port and two port *active* networks; examples of open-circuit and short-circuit stable/unstable *active* circuits
Unit 3	Fundamentals of Graph theory: concepts of graph, tree, cut-set, fundamental cut-sets, fundamental loops; formulation of linearly independent node and loop equations of *active*-RC circuits; Tellegen Theorem, its implications and applications; adjoint and dual active networks
Unit 5	State variable analysis and Synthesis: formulation of state equations of passive and *active* circuits, state transition matrix and its properties, solution of state equations, One port passive and *active* synthesis; state variable synthesis of voltage-ration and current ratio transfer functions using *active* elements

Text Books:

[T1] Active Network Analysis', Wai Kai Chen, World Scientific, 1991

[T2] RC Active Circuits`, L. T. Bruton, Prentice Hall Inc. New Jersey, 1980

[T3] Fundamentals of Network Analysis and Synthesis`, Behrouz Peikari, Jaico Publishing House, 2008

[T4] Analysis and Synthesis of Linear Active Networks`, S. K. Mitra, John Wiley and Sons Inc. 1968

Course No.	Title of the Course	Credits	Course Structure	Pre-Requisite
ECECC02	Electronic Devices and Circuits	4	L-T-P 3-0-2	None

COURSE OUTCOMES (CO)

After completing the course, the students should be able to:

CO 1: Acquire basic knowledge of the physical structures and operation of various electronic devices

CO 2: Be familiar with the physical structure and operation of BJT, FET and MOSFET

CO 3: Understand the concept of load line, biasing of BJT, FET and MOSFET

CO 4: Develop the capability of analyzing amplifier circuits using BJT, FET and MOSFET

CO 5: Explore the applications of BJTs and MOSFETs as Current Sources and Sinks

CO:6 Acquire the knowledge of operation and applications of various Optoelectronic devices

Unit No.	Topics
Unit 1	Review of semiconductor diodes, Mass action law, carrier concentrations, Graded and step graded semiconductors, calculation of barrier potential, Drift and diffusion currents; Physical structure and operation of Zener Diode, Schottky diode, Varactor diode, Step recovery diode
Unit 2	Physical structure and modes of operation of BJT, input, output and transfer characteristics, The Ebers-Moll model for BJT; Biasing schemes for BJT, determination of operating point; bias stability and bias stabilization.
Unit 3	BJT as an amplifier and switch (NPN and PNP both); Various configurations: CE, CB and CC; Low frequency transistor model, Small signal analysis, Estimation of voltage gain, input resistance, output resistance; simple current mirror, Bipolar current sources/sinks and bandgap references.
Unit 4	JFET characteristics and working principle, Biasing schemes for JFET, Small signal analysis; JFET amplifier, JFET as a voltage-controlled resistance (VCR); MOSFET operation; enhancement mode and depletion mode; Biasing schemes for MOSFET, various configurations: CD, CS, CG; MOSFET as a VCR; Complimentary MOS (CMOS), CMOS digital inverter; Simple Current mirror, MOS current sources/sinks
Unit 5	LED, photo-diode, opto-coupler, opto-isolator, photo transistor; Power electronic Devices: Thyristor, UJT, SCR.
	References: **[T1]** Sedra, Adel S. and Smith, K. C., *Microelectronic circuits*. New York: Oxford University Press, 1998. **[T2]** Boylestad, Robert L. and Louis Nashelsky, *Electronic Devices and Circuit Theory*. Pearson Education, India, 2009. **[T3]** Millman, Jacob, and Arvin Grabel. *Microelectronics*. McGraw-Hill, Inc., 1987. **[T4]** Malvino, Albert, and David Bates. *Electronic Principles with Simulation CD*. McGraw-Hill, Inc., 2006. **[T5]** David A. Bell *Electronic Devices and Circuits*, Oxford University Press, Fifth edition.
	List of Experiments **1.** Plot the input and output characteristics of BJT in CE configuration **2.** Design CE amplifier using BJT: measure input resistance, voltage gain and plot the frequency response curve and measure bandwidth.

	3. Design CB amplifier using BJT: measure input resistance, voltage gain and plot the frequency response and measure bandwidth. **4.** Design CC amplifier using BJT and find input and output resistance, voltage gain and plot the frequency response curve and measure bandwidth. **5.** To perform the operation of BJT as switch (both npn and pnp) **6.** To determine the input and output characteristics of JFET **7.** Design RC coupled JFET Amplifier: measure input resistance voltage gain and plot the frequency response and measure bandwidth **8.** To perform and plot the characteristics curve I_D versus V_{DS} of MOSFET using CD4007 and measure the various parameters of the MOSFET. **9.** Design common source amplifier using CD4007, measure its voltage gain and plot frequency response and measure bandwidth **10.** To Study the operation of UJT as a Relaxation Oscillator **11.** To draw the V-I Characteristics of Silicon controlled rectifier and measure latching and holding current **12.** Minor project based upon the use of LED, photo-diode, opto-coupler, opto-isolator, photo transistor

Course Code	Type	Subject	L	T	P	Credits	CA	MS	ES	PCA	PES	Pre-Requisites
ECITC03	CC	Data Structures and Algorithms	3	0	2	4	15	15	40	15	15	None

COURSE OUTCOMES (CO)
By the end of the course students will be able to:

CO1: Understand and remember algorithms and its analysis procedure.
CO2: Illustrate the concept of data structures through ADT including List, Stack, Queues.
CO3: To design, implement and compare various data structure algorithms.
CO4: To select appropriate data structure for representation of the data in the real world.
CO5: To formulate new solutions or improve existing applications using data structure algorithms.

COURSE CONTENT No. of hours:36

Unit 1
Introduction to data structure: Arrays, Stacks and Queues: Fundamentals and Representations, Applications of
Arrays, Stacks and Queues, Sparse Matrices, Space and time complexity of algorithms.
[6hrs]

Unit 2
Linked lists: Singly/Linear Linked lists, Linked list implementation of Stacks and Queues, Doubly

and Circular
Linked Lists, Applications.

[6hrs]

Unit 3
Trees: Binary Trees, B-Trees, N-ary Trees, B+Trees, Tree Traversals, infix to post fix conversion, construction of tree from preorder and inorder notation, Storage of Trees, Threaded trees, AVL trees, Red and Black trees, Application of Trees, Hashing, Heaps: Types of heaps, minheap, maxheap, double heap.

[8 hrs]

Unit 4
Graphs: Types, Terminology and Representations, Graph Traversals, Applications of Graphs.

[8 hrs]
Unit 5
Searching and Sorting: Sequential and Binary Searching, Search trees, Sorting Techniques.

[8 hrs]

SUGGESTED READINGS:

1. E. Horowitz and S. Sahani, "Fundamentals of Data Structures in C", 2nd Edition, Universities Press.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein "Introduction to Algorithms", 3rd Edition, The MIT Press Cambridge , Massachusetts London, England McGraw-Hill Book Company.
3. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", Addison-Wesley.
4. Schaum's Outline Series, "Data Structure", TMH, Special Indian Ed., Seventeenth Reprint.
5. Y. Langsam et. al., "Data Structures using C and C++",PHI
6. N. Dale and S.C. Lilly, D.C. Heath and Co., "Data Structures"
7. R. S. Salaria, Khanna, "Data Structure & Algorithms", Book Publishing Co. (P)Ltd..
8. Richard F. Gilberg and Behrouz A. Forouzan, "Data Structure A Pseudocode Approach with C", Cengag Learning, 2nd Ed.,

4.3.3 SYLLABI OF PROGRAM CORE COURSES : III SEMESTER

Course No.	Title of the Course	Course Structure	Pre-Requisite
ECMTC04	Mathematics for Signal Processing and Communication	3L-1T-0P	None

COURSE OUTCOMES (CO)

CO1: Know the concepts of Group theory, homomorphism and isomorphism and its applications, Rings
CO2: Know the concepts of Vector Spaces and Linear Transformations
CO3: To know the concepts of vector calculus such as gradient, curl, divergence and integral theorems such as Green's Theorem, Stoke's Theorem and Gauss Divergence Theorem and their applications in various fields
CO4: To know real integrals the concepts of functions of complex variables and its applications to evaluate

CO5: To know evolution of Partial Differential Equations and its methods of solutions for real life problems

COURSE CONTENT:

Abstract Algebra: Groups, definitions and examples, Abelian Group, Homomorphisms, Subgroups, Order of a Group, Cyclic Groups, Factor Groups, Simple Groups and Composition Series, Abelian Groups, Rings: Definitions and Examples, Homomorphisms, Integral domain, Fields.

Linear Algebra: Vector space, Vector spaces over Q, R and C, Subspaces, linearly dependent and linearly independence set of vectors, Basis and dimension of a vector space, vectors in plane and space, Linear Transformation.

Vector Calculus: Differentiation of a vector function, scalar and vector fields, Gradient, Divergence, Curl, line integral, independence of path, Green's theorem and applications. Surface Integral, Stoke's theorem and applications; Volume Integrals, Gauss Divergence theorem and applications.

Complex Variables: Functions of a complex variable, analytic functions, harmonic functions, Cauchy-Riemann equations (Cartesian and polar form). Linear fractional transformation, Conformal mapping, Mapping of elementary functions (exponential, trigonometric, hyperbolic and logarithm functions), Contour integration, Cauchy's integral theorem and formula, zeroes, Singularities, Poles, Residue theorem, Evaluation of real integrals (around unit circle, no singularity on real line, and singularity on real line).

Partial Differential Equations: Solution of first order equations- Lagrange, nonlinear first order, Charpit's method, higher order linear equations with constant coefficients. Separation of variables, Solutions of Heat and Wave equations (one dimensional only), Laplace equation.

Recommended Books:

1. I. N. Herstein, ``Topics in Algebra, '' Wiley Publishing
2. J. B. Fraleigh, ``A First Course in Algebra,'' Narosa Publication
3. Jain and Iyenger, ``Advanced Engineering Mathematics,'' Narosa Publication
4. Kreyszig, ``Advanced Engineering Mathematics,'' Wiley Publication
5. Greenberg, ``Advanced Engineering Mathematics,'' Pearson Education

Course Code	Course Name	Course Structure	
ECECC05	Signal and Systems	3-1-0	L-T-P

COURSE OUTCOME (CO):

CO1: To understand the concept of signals and systems and their classifications.

CO2: To understand the concept of convolution and its applications.

CO3: To analyze the spectral characteristics of continuous-time and discrete-time periodic and aperiodic *signals* using Fourier analysis.
CO4: To analyze continuous-time LTI systems in the time domain and s-domain
CO5: To analyze discrete-time LTI systems in the time domain and z-domain

COURSE CONTENT:

Unit-I:

Concept definition & classification of signals– Continuous time (CT) and Discrete Time (DT) signals, Standard signals- Step, Ramp, Pulse, Impulse, Real and complex exponentials and Sinusoids, Properties of unit impulse in continuous and discrete domain, Concept of orthogonality, Transformation of independent variables, Classification of systems & their properties, Waveform synthesis.

Unit-II:

Convolution integrals and its properties, Properties of LTI System, impulse response, step response, convolution sum and its properties, graphical method.

Unit-III:

Fourier series representation of continuous time and discrete time signals and their properties. Magnitude and Phase spectrum of signals.

Continuous time Fourier Transform and its properties, Discrete time Fourier Transform and its properties. Magnitude and Phase representations of frequency response of LTI systems. Introduction to LP/BP/HP/All Pass systems. Relationship between discrete time signals and Sampled continuous time low pass signals.

Unit-IV:

Laplace Transform and its properties, concept of ROC and its properties. Inverse-Laplace transform. Unilateral Laplace transform and its applications. Computation of impulse response, total response (zero state and zero input response) & transfer function using Laplace transform. Stability and causality of continuous-time LTI Systems.

Unit-V:

Z-Transform and its properties, ROC and its properties, Inverse Z-transform; Power series expansion and partial fraction expansion. Relationship between Z-transform, Fourier transform and Laplace transform. Stability and causality of discrete-time LTI System. Unilateral Z transform and its applications. Computation of impulse response, total response (Zero state and Zero input response) & Transfer function using Z-Transform.

Text Books:

[T1] Alan V. Oppenheim, Alan S. Willsky, S. Hamid Nawab, Signals and Systems Prentice Hall India, 2nd Edition, 2009.

Reference Books:

[R1] Simon Haykin and Barry Van Veen, "Signals and Systems", John Willey & Sons, Inc, second edition 2013.

[R2] Tarun Kumar Rawat, Signals and Systems, Oxford University Press, 1st Edition, 2010.

Course Code	Course Name	Course Structure	
ECECC06	Probability Theory and Random Process	L-T-P	3-1-0

COURSE OUTCOME (CO):

CO1: To understand the probability theory and random variables for the analysis of random phenomena.
CO2: To characterize probability models and function of random variables based on single & multiples random variables.
CO3: To evaluate and apply moments & characteristic functions and understand the concept of inequalities.
CO4: To understand the concept of random processes and determine covariance and spectral density of stationary random processes.
CO5: To understand representation of low pass and band pass noise models.

COURSE CONTENT:

Unit-I:

Probability Theory: Introduction to Random Variables, PDF, CDF. Continuous/Discrete/Mixed type random variables. Normal, Exponential, chi-square, Rayleigh, Nakagami-m, uniform distributions, etc. Bernoulli, Binomial, Poisson distributions. Expectation, Variance, Moments, Characteristic functions.

Unit-II:

Joint Random variables and random vectors. Independent, uncorrelated random variables. Joint Expectation, co-variance, Joint Moments, joint characteristic functions, conditional distributions, conditional expectation. Multidimensional Gaussian law

Unit-III:

Functions of random variables of the type Y=g(X), Z=g(X,Y) and two functions of two random variables, Inequalities of Chebyshev and Schwartz. Chernoff Bounds.

Unit-IV:

Random processes. First-order stationary processes, second order and wide sense stationary process, Nth order and strict sense stationary process, Time averages and ergodicity, Mean ergodic process, Auto correlation function and its properties, Cross- correlation function and its properties, Covariance functions and their properties, discrete time processes and sequences,

Unit-V:

Power density spectrum and its properties, Linear systems with random inputs. Random signal response, Auto correlation functions of the response, Cross correlation functions of input and output system, Power density spectrum of the response, Spectral representation. Wiener- Khinchine theorem. Gaussian process, White process, Response of LTI systems to White Noise

Text Books:

[T1] H. Stark and J. W. Woods, Probability and Random Processes with Applications to Signal Processing, 3rd Edition Prentice Hall.

Reference Books:

[R1] A Papoulis and S. Pillai, Probability, Random Variables and Stochastic Processes, 4th Edition, PHI, 2002.
[R2] P.Z. Peebles, Probability, Random Variables and Random Signals, 4th Edition, TMH, 2001.

Course Code	Course Name	Course Structure		Pre-Requisite
ECECC07	Microelectronics	L-T-P	3-0-2	EEE

COURSE OUTCOME (CO):

After completing the course, the students should be able to:

CO-1 Understand the modelling and analysis of single-stage MOS and BJT amplifiers
CO-2 Analyse differential and multi-stage amplifiers using BJTs and MOSFETs
CO-3 Understand the basic parameters and analysis of bipolar and CMOS op-amps
CO-4 Understand the basic negative feedback amplifier topologies and their characteristics
CO-5 Understand the importance of positive feedback and the realisation of sinusoidal oscillators
CO-6 Understand power amplifiers realised from power BJT and MOSFETs and IC power amplifiers

Unit No.	Topics
Unit 1	Review of Small signal operation and models for MOSFET and BJT, high frequency models of MOSFET and BJT, Single stage integrated circuit (IC) amplifiers, Widlar, Wilson and Cascode Current mirrors, IC biasing, High frequency response of CS and CE amplifiers with passive and active loads, high frequency response of CG and CB amplifiers with active loads, CS and CE amplifiers with source de-generation, Source follower and emitter follower, Cascode amplifier.
Unit 2	Differential and multi stage amplifiers: MOS differential pair and its small signal operation. BJT differential pair, Non ideal characteristics of differential amplifiers, Differential amplifiers with active load; Multi-stage amplifiers using MOSFETs and BJTs.
Unit 3	Feedback amplifiers and Sinusoidal oscillators: General feedback structure, Properties of negative feedback, Four basic feedback topologies, Loop gain, Stability problem, Basic principle of sinusoidal oscillators, RC phase shift and Wien bridge oscillators, LC and crystal oscillators. The
Unit 4	Typical architectures of internally-compensated-type Bipolar and CMOS op-amps: Input common mode range and output swing, DC voltage gain, CMRR, Frequency response, phase margin and gain margin, Slew rate and unity gain bandwidth, Power supply rejection ratio.
Unit 5	Output stages and power amplifiers: Class A, Class B, Class AB output amplifier stages, Class C amplifiers, Power BJTs, MOS power transistor, Discrete and IC power amplifiers.

References:

1. Adel S. Sedra and K. C. Smith, *Microelectronic Circuits,* New York: Oxford University Press, 1998.
2. P. R. Gray, P. J. Hurst, S. H. Lewis Meyer, *Analysis and Design of Analog Integrated Circuits, Fourth Edition 20*09.
3. Jacob Millman and Arvin Grabel. *Microelectronics,* McGraw-Hill, Inc., 1987.

List of Experiments:

1. Evaluation of the performance of BJT Wilson Current mirror
2. Evaluation of the performance of MOS Cascode Current mirror
3. Evaluation of the performance of a Cascode amplifier
4. BJT-based differential amplifier with active load (and current source biasing)
5. MOS differential amplifier
6. Evaluation of the performance of a two-stage CMOS op-amp

7. Study of the properties of negative feedback amplifiers (implementation of series-series, series-parallel, parallel-series and parallel-parallel topologies)
8. Study of class AB push-pull amplifier
9. BJT-based RC phase shift oscillator
10. Study of Colpitts and Hartley Oscillators

Note: Course teachers may design 3-4 new experiments/small projects in addition to the above suggested practical exercises.

Course Code	Course Name	Course Structure		Pre-requisites
ECECC08	Digital Circuits and Systems	L-T-P	3-0-2	EEE

COURSE OUTCOMES

After the completion of the course, the students should be able to:

CO1: Get familiarized with number systems, codes, logic gates and Boolean algebra
CO 2: Understand the properties and operation of contemporary logic families.
CO 3: Design and implement basic combinational and sequential circuits.
CO 4: Understand the motivation and need for modelling and synthesis using Hardware Description Languages.
CO 5: Understand system design processes using programmable logic devices, operation of building blocks of a digital computer system.

UNIT-1: Logic functions and operation of logic families

Digital design representations (e.g., truth table, equations, schematic, etc.), Different codes and their properties (e.g., Gray, BCD, 1-Hot etc.). Signed and unsigned Number representation and conversion. Binary addition and subtraction (unsigned and one's and two's complement). Boolean equations and terminology (SOP, POS, minterm, maxterm), Logic implementations of equations (2-level, AND/OR, NAND/NOR, etc.). Characteristics of TTL and CMOS logic families (i.e., logic thresholds, delay, Noise margin, fan-in and fan-out, power dissipation), TTL to CMOS and CMOS to TTL interfacing.

UNIT-2: Elements of Combination logic

Overview of Boolean algebra, Simplification of logic functions using (i) K-map up to 5 variables with don't care conditions (ii) Tabulation (Quine-McCluskey) method. Combinational logic design: half and full adder, half and full subtractor. Multiplexers, Encoders, Decoders, Priority encoders, Comparators, Magnitude comparator, Demultiplexer. Code converters: BCD to Seven Segment Display (SSD), Binary to SSD. Introduction to a PROM and its use for implementing arbitrary logic functions.

UNIT-3: Elements of sequential logic.

SR Latch, D Latch, Master-Slave operation of a latch. Latch versus flip-flop, Flip-Flops (D, SR, JK, and T), Registers, Shift registers (SISO, SIPO, PISO), Linear feedback shift registers. Ripple and synchronous counters. PWM generator. Binary multiplier. Synchronous and asynchronous sequential circuits. Mealy and Moore sequential models. Design of Finite state machine using state table, state diagram, state

assignment (binary, Gray, one-hot), next state and output equations. Example FSM implementations.

UNIT-4: Hardware Description Language

Modern digital design methods. Motivation and need for hardware description language. Introduction to VHDL and important terminology. VHDL Library and packages. Entities, architecture and configurations. Signals and data types. Operators and processes. VHDL code examples of all combinational and sequential circuits from unit-2 and unit-3 including FSM.

UNIT-5: Digital System Design and Programmable Logic Devices

Digital system design using FSMD approach. Design of illustrative systems using VHDL, such as game of dual-dice, rock-paper-scissor, reaction timer, period counter etc. Elements of CPU - RAM and DRAM; ALU; control logic using FSM and microprogram control approach; register file. Introduction to programmable logic devices - PAL/PLA, GAL, CPLD and FPGA. Introduction to JTAG.

SUGGESTED READINGS

1. Donald D. Givone, "Digital Principles & Design'', Tata McGraw Hill. ISBN: 978-0070529069.
2. M.M. Mano, M.D. Cilleti, "Digital Design'', 6th edition, Pearson, 2018. ISBN: 978-9353062019
3. C. Roth, L. Kinney, "Fundamentals of Logic Design ", 7th Edition Cengage Learning, 2014. ISBN: 978-8131526156.
4. Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital Logic with VHDL Design", McGraw Hill. ISBN: 978-1259025976.
5. Pong P. Chu, "FPGA Prototyping by VHDL" Wiley. ISBN: 978-1119282747

LIST OF EXPERIMENTS

1. Characterization of Logic Family (Use 74HC/LS/HCT04 on a breadboard)
a. Find out logic threshold values and noise margins.
b. Propagation Delay time measurement of inverter using ring oscillator with 3, 5, 7, 9 and 11 inverter elements.
2. Design, Implement and verify a: half-adder, (two numbers of 2-bits each) full-adder, half-subtractor, (two numbers of 2-bits each) Full-subtractor using Xilinx Design Suite (Schematic capture) and verify on a CPLD/FPGA Board
3. Design, implement and verify a 2-bit binary comparator using basic gates. Use unsigned binary numbers as inputs. The comparator should have three outputs – Equal, Greater than and Less than.
4. Design, Implement and verify a 3x8 decoder and 8x1 multiplexor using basic gates.
5. Design, implement and verify a circuit to implement any combinational circuit with 4-inputs using a 16x1 multiplexor library symbol in the Xilinx design suite library.
6. Design, implement and verify a using a 16x1 multiplexer: a BCD to Seven Segment Decoder and a Binary to Seven Segment decoder.

7. Design, implement and verify a 4-bit binary counter and a decimal counter using basic gates and DFFs. Demonstrate the output on LEDs.
8. Design, implement and verify using FSM, a D type flip-flop with an enable signal. If enable is '1', the flip flop captures the input signal, if enable is '0', the flip flop retains the previous value.
9. Design, implement and verify, using FSM approach, a 3-bit up-counter with an enable signal. If enable is '1', the counter counts normally. If enable is '0', the counter retains the last value.
10. Implement an electronic dice using FSM approach. The dice should display random numbers between 1 and 6. The input to the dice is an input 'throw dice' signal, apart from the clock.

Note:

1. Experiments 2-10 will be implemented on a CPLD/FPGA board.
2. Course teachers may design 3-4 new experiments/small projects in addition to the above suggested practical exercises.

4.3.4 SYLLABI OF PROGRAM CORE COURSES : IV SEMESTER

Course Code	Course Name	Course Structure		Pre-requisite
ECCOC09	Machine Learning and Artificial Intelligence	L-T-P	3-0-2	Statistics, Calculus, Linear Algebra, Probability, Programming Knowledge

COURSE OUTCOMES

CO1: Be able to develop an understanding of the fundamentals of machine learning and statistical pattern recognition.

CO2: Be able to understand the fundamentals of artificial intelligence, learning systems, their goal and applications.

CO3: Be able to gain an insight into the various components of machine learning such as supervised learning, unsupervised learning, learning theory, reinforcement learning and adaptive control.

CO4: Be able to understand and implement various machine learning algorithms and AI heuristic techniques.

CO5: Acquire skills that can be applied to various applications like robotic control, data mining, autonomous navigation, bioinformatics, speech recognition, and text and web data processing.

CONTENTS

UNIT-I

Introduction: Definition of learning systems, Goals and applications of machine learning.

Types of ML: Supervised, Unsupervised and Reinforcement learning

Experimental Evaluation of Learning Algorithms: Measuring the accuracy of learned hypotheses, Bias, variance, comparing learning algorithms: cross-validation, learning curves, and statistical hypothesis testing.

UNIT-II

Machine learning algorithms

Supervised learning algorithms: Classification, Regression Nearest neighbor, Naïve bayes, Decision Trees, Linear regression, Support Vector Machines, Neural networks

Unsupervised learning algorithms: Clustering, Association, K-means

Reinforcement Learning: Q-learning, Temporal Difference, Deep adversarial networks

UNIT-III

Introduction: Introduction to AI, Turing test

Reasoning, Logic & Theorem Proving: Review prepositional & predicate Calculus, Resolution

Unit –IV

Heuristics Searches: Best first search, depth first search, Hill Climbing, A*, AO*, Constraint Satisfaction, Means-ends analysis

Learning: Rote Learning, Learning by taking advice, Explanation based learning, Discovery, Analogy

UNIT-V

Case studies and Applications of AI and ML for real world problems

Practical Outline:
Implementation of ML algorithms: Nearest neighbour, Naïve bayes, Decision Trees, Linear regression, Support Vector Machines, Neural networks, K-means, Q-learning
Implementation of heuristics techniques: Best first search, depth first search, Hill Climbing, A*, AO*, Constraint Satisfaction, Means-ends analysis

TEXT BOOKS

1. Tom Mitchell, Machine Learning. McGraw-Hill
2. Elaine Rich and Kevin Knight, Artificial Intelligence, Tata. McGraw-Hill

REFERENCE BOOKS

1. Richard Duda, Peter Hart and David Stork, Pattern Classification, 2nd ed. John Wiley & Sons, 2001.
2. Richard Sutton and Andrew Barto, Reinforcement Learning: An introduction. MIT Press, 1998
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, The Elements of Statistical Learning. Springer, 2009
4. Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig

"Artificial Intelligence: A New Sythesis" by Nils J Nilsson.

Course Code	Course Name	Course Structure	
ECECC10	Electronic Design Workshop	L-T-P	2-0-4

Course Objectives:

1. To be able to visualize a system/product in terms of hardware and software building blocks using a project based learning approach.
2. Learn useful mechanical and electronic fabrication processes.
3. Learn necessary skills to build useful and standalone system/project with enclosures
4. Learn necessary skills to create print and electronic documentation for the system/project
5. Build, test and document a useful power supply before the mid term.
6. Build, test and document a useful project/system.
7. Learn various debugging tools and techniques.

Unit #	Topics
1	Electronic component familiarization, Understanding electronic system design flow - visualization of system requirements, time planning using Gantt chart, schematic design and PCB layout, circuit fabrication, soldering and testing, heat sink and cooling for critical components, system wiring, building system enclosure, system testing and debugging, documentation. Documentation using Google docs, Overleaf. Documentation from firmware - Doxygen. Version control tools for

	documentation as well as firmware release management - GIT and GitHub. Familiarization and use of basic measurement instruments - DSO including various triggering modes, 1X and 10X probes, DMM (including CAT I, CAT II, CAT III, and CAT IV type of DMMs), LCR bridge, Signal and function generator. Logic analyzer or MSO. Bench power supply (with 4-wire output)
2	Circuit prototyping using (a) breadboard, (b) Zero PCB (c) 'Manhattan' style and (d) custom PCB. Schematic design and PCB layout using Eagle CAD. Gerber creation and 3D visualization for fitting. Single, double and multilayer PCBs. Single- and double-sided PCB prototype fabrication in the lab. Soldering using soldering iron/station. Soldering using a temperature controlled reflow oven. Automated circuit assembly and soldering using pick and place machines.
3.	Electronic circuit building blocks including common sensors. Arduino programming and use. Getting acquainted with the Arduino IDE and Basic Sketch structure. Digital Input and output. Measuring time and events. PWM. Serial communication. Analog input. Interrupts programming.
4.	Power sources and power supply regulator design. Linear and switching power supply. Transformerless power supply. Zener regulator, op-amp based regulator. 723 IC regulator. 3-terminal IC regulators. LDO and micropower regulators. Buck, Boost and Buck-boost switching regulators. Short circuit and over-voltage protection. Wireless power supply. USB-PD. Battery types and characteristics. Battery charging for lead-acid and lithium type batteries.
5	Mechanical fabrication processes - 3-axis CNC, 3D printing, mini lathe, drilling, Laser cutting, Laser engraving etc. 3D printing technology - FDM, SLS and SLA. 3-axis CNC operation. 2D file designing for drilling, cutting, milling, Laser Cutting and engraving using Flatcam, Inkspace, OpenBSP. 3D file designing using Sketchup, FreeCAD, Prusa Slicer.

Laboratory Activities:

S. No.	List of Lab activities and experiments
1.	Electronic component familiarization using Charts (with proper samples) for package familiarization for TH, SMD type components both active and passive components (R, L, C, ICs, BJT and MOSFETs, Switches, Connectors, potentiometers, opto-coupler, opto-isolator, optical fibres, LDR, photocell, photo-transistor, laser diodes, LEDs, etc). Understanding various types of resettable and non-resettable fuses, EMI filters and MOV varistors. Visit to local electronics markets.
2.	Learning to use DSO properly with normal and single triggering modes using non-periodic signals. Estimating the input impedance of a DSO. Learning the use of 10X probe using high-frequency or high-slew rate signals. Measuring input impedance of DMM. Measuring very low

	resistance using 4-probe methods. Measuring output impedance of signal generator, power supply, battery etc.
3.	Simple circuit fabrication using (a) breadboard (b) Zero PCB (c) Manhattan style PCB.
4.	3D printing small object or enclosure. CNC machining of stock copper clad PCBs. Laser cutting of Acrylic sheets, laser engraving on Acrylic sheet.
5.	PCB layout and fabrication in lab using PCB milling machine and/or toner transfer method. Single- and double-sided PCB fabrication. Identification layer transfer on PCB.
6.	Experiments with various types of batteries and small solar panels.
7.	Design, fabrication and testing of a linear or switching power supply for using the PCB fabrication techniques, for use in mini-project.
8.	Stand alone mini-project using Arduino (or any other microcontroller of your choice), some sensors, displays etc. The mini project should have its own power supply or battery power source. The mini-project should be properly enclosed in lab designed and fabricated enclosure which could be 3D printed or made using CNC machining. The mini-project would have a mandatory documentation report.

Reference Books:

S.No	Title
1.	*The Art of Electronics.* 3rd edition. Paul Horowitz and Winfield Hill. Cambridge University Press. ISBN: 9780521809269
2.	*Practical Electronics for Inventors.* 4th edition. Paul Sherz and Simon Monk. McGraw Hill. ISBN-13: 978-1259587542
3.	*Encyclopedia of Electronic Components (Volume 1, 2 and 3).* Charles Platt. Shroff Publishers. ISBN-13: 978-9352131945, 978-9352131952, 978-9352133703
4.	*Building Scientific Apparatus.* 4th edition. John H. Moore, Christopher C. Davis, Michael A. Coplan and Sandra C. Greer. Cambridge University Press. ISBN-13: 978-0521878586
5.	*Programming Arduino: Getting Started With Sketches.* 2nd edition. Simon Monk. McGraw Hill. ISBN-13: 978-1259641633

6.	*Debugging.* David J. Agans. Amacom. ISBN: 0814474578
7.	*Make Your Own PCBs with EAGLE: From Schematic Designs to Finished Boards.* Simon Monk and Duncan Amos. McGraw Hill Education. ISBN-13 : 978-1260019193.
8.	*Pro GIT.* 2nd edition. Scott Chacon and Ben Straub. Apress. ISBN-13 : 978-1484200773

NSUT
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

Course Code	Course Name	Course Structure	
ECECC11	Communication Engineering	L-T-P	3-0-2

COURSE OUTCOME (CO):

CO1: To Gain the knowledge of components of analog communication system.
CO2: To understand the generation, transmission and reception of AM signals.
CO3: To understand the generation, transmission and reception of FM signals.
CO4: To evaluate the performance of AM/FM receivers in the presence of white noise
CO5: To understand the pulse modulation techniques and basic baseband digital communication system.

COURSE CONTENT:

Unit-I:

Introduction to communication system, Communication Channels, Electromagnetic spectrum, Need for modulation, Distortion less transmission; Hilbert transform; Pre-envelope, complex envelope and canonical representation of band pass signals.

Unit-II:

Definition, Time domain and frequency domain description - AM, DSB-SC, single tone modulation, power relations in AM waves, Generation of AM waves (square law, Switching), Envelop detector, Generation of DSB-SC Waves (Balanced, Ring), Coherent detection of DSB-SC Modulated waves, COSTAS Loop, SSB Modulation: Time domain, Frequency domain description, Frequency discrimination, Demodulation of SSB Waves, Frequency Division Multiplexing, Vestigial sideband modulation and demodulation, I-Q representation of all modulation formats. Comparison of AM Techniques. AM broadcasting Techniques: Super heterodyne Receivers.

Unit-III:

Basic concepts, Frequency and Phase Modulation, Relation between phase and frequency modulation, Single tone frequency modulation, Spectrum Analysis of Sinusoidal FM Wave, Narrow band FM, Wide band FM, Constant Average Power, Transmission bandwidth of FM Wave, Generation of FM Waves. FM demodulation (Slope detector, PLL). FM broadcasting Techniques.

Unit-IV:

Representation of Band Pass white Noise, Thermal Noise, Noise equivalent temperature, Noise figure of cascaded communication System, Equivalent Noise BW, Effect of White Noise on Amplitude Modulation/Demodulation systems: SNR/FOM calculations for coherent and envelope detection. Effect of Noise on Angle Modulation – Signal to Noise Ratio; Threshold Effect; Pre emphasis & De-emphasis in FM.

Unit-V:

Sampling Theorem, Sampling of Low Pass Signals, Reconstruction of Sampled Signals, Interpolation, Aliasing Effects, Aperture Effect; Requirements of a line encoding format, various line encoding formats- Unipolar, Polar, Bipolar etc. and their spectrum. PAM, PWM, PPM, Modulation and Demodulation; Pulse Code Modulation (PCM), TDM, Quantization Noise, Optimal Linear Prediction, Differential PCM (DCPM), Delta Modulation, Adaptive DM.

Text Books:

[T1] Simon Haykin, Communication Systems, 5th Edition John Wiley.

Reference Books:

[R1] JG Proakis, Communication Systems, 2nd Edition, Pearson Education.

[R2] B.P. Lathi and Zhi Ding, Modern Digital and Analog Communication Systems, 4th Edition, Oxford University Press.

Course Code	Course Name	Course Structure	
ECECC12	Microprocessors and Computer Architecture	L-T-P	3-0-2

COURSE OUTCOME (CO):

CO1: To comprehend the instruction set architecture of 8085 microprocessor, instruction cycle and concept of interfacing

CO2: To understand instruction set architecture of 8086 microprocessor.

CO3: To familiarize students with processor architecture, instruction set architecture and assembly language programming in general

CO4: To develop understanding and operation of memory system and different memory types.

UNIT-1

8086 PROCESSOR: Von-Neumann & Harvard CPU architecture and CISC & RISC CPU architecture 8086 CPU Architecture. Addressing modes.

Physical memory Organization, General Bus operation cycle, I/O addressing capability, Special processor activities, Minimum mode 8086 system and Timing diagrams, Maximum Mode 8086 system and Timing diagrams.

INSTRUCTION SET OF 8086: Data transfer and arithmetic instructions. Control/Branch Instructions, Logical Instructions, String manipulation instructions, Flag manipulation and Processor control instructions, Illustration of these instructions with example programs. Assembler Directives and Operators, Assembly Language Programming and example programs

UNIT-2

Introduction to stack, Stack structure of 8086, Programming for Stack. Interrupts and Interrupt Service routines, Interrupt cycle of 8086, NMI, INTR, Interrupt programming, Passing parameters to procedures, Macros, Timing and Delays.

Basic Peripherals and their Interfacing with 8086: Static RAM Interfacing with 8086 (5.1.1), Interfacing I/O ports, PIO 8255, Modes of operation – Mode-0 and BSR Mode, Interfacing ADC-0808/0809, DAC-0800, 8259.

UNIT-3

CPU structure and functions, processor organization, ALU, datapaths, internal registers, status flags; System bus structure: Data, address and control buses, Processor control, micro-operations, instruction fetch, hardwired vs. microprogrammed control, microinstruction sequencing and execution,

Unit-4

Instruction set principles, machine instructions, types of operations and operands, encoding an instruction set, assembly language programming, addressing modes and formats. Pipelining: basic concepts of pipelining, throughput and speed, pipeline hazards, Comparison between CISC and RISC architecture, Introduction to RISC-V and

domain specific architectures
UNIT-5
Memory system, internal and external memory, memory hierarchy, cache memory and its working, virtual memory concept, I/O organization; I/O techniques: interrupts, polling, DMA; Synchronous vs. asynchronous I/O.

Text Books:

[T2] Hall, D.V., "Microprocessors and Interfacing", 2nd Ed., Tata McGrawHill. 2006
[T3] D.A. Patterson, J.L. Hennesey, "Computer Architecture: A Quantitative Approach", Sixth Edition, Morgan Kaufmann, 2017.
[T4] Brey, B.B., "The Intel Microprocessors", 6th Ed., Pearson Education. 2003
[T5] Mano, M.M., "Computer System Architecture" 3rd Ed., Prentice-Hall of India. 2004
[T6] Mano M. M., Cilleti M. D., " Digital Design"5th Edition, Pearson 2013
[T7] Rajaraman, V. and Radhakrishnan, T., "Computer Organization and Architecture", Prentice-Hall of India. 2007
[T8] Govindrajalu, B., "Computer Architecture and Organization", Tata McGraw-Hill. 2004
[T9] Stallings, W., "Computer Organization and Architecture", 5th Ed., Pearson Education. 2001
[T10] John E. Uffenbeck," 8086/8088 Design programming and interfacing," Prentice Hall of India 1997.

List of Experiments

Basic Exercise 1:
To understand the operation of the kit, the various components on the kit and their functions. List out the major components that you see on the kit. Find out the function of these components and classify these components into the various broad categories of components Write a suitable report.

Basic Exercise 2:
Read the manual supplied with the Vinytics kit. The keyboard on the kit has two groups of keys: red keys and black keys. Understand the operation of all the red keys. The red keys are function keys and the black keys are data keys.
Power up the kit and using the function keys store a few arbitrary numbers in the SRAM chip of the kit. Now power off the kit for a few minutes and power it up again. Inspect the SRAM locations again and verify that the contents of these SRAM locations have not changed even after you powered down the Kit. Find out why. Give your suggestions for the implementation of a circuit that would help retain the contents of the SRAM even if the main power is turned off. Can you replace the SRAM chips with any other type of chip and still be able to modify the contents of those chips? What would be the advantages or disadvantages of those alternatives? Write a report.

Basic Exercise 3:

At this point you are familiar with the basic operation of the kit..
Write a program to add a series of 10 numbers. These numbers are stored in consecutive memory locations in the SRAM. The result should be stored in the memory locations following the input data. What should be the size of the result location? What if you were to add a series of 1000 numbers? How much result storage space would be sufficient? Write a report.

Experiments:

1. Write a program to arrange the 10 numbers in SRAM memory in ascending order. Repeat you experiment to arrange the data in descending order.
2. Develop a subroutine for Multiply and divide operations.
3. Write routines to convert Binary to ASCII, ASCII to binary, binary to BCD, BCD to binary.
4. Write a program to test the RAM on the Kit.
5. Study the operation of 8255 Interface Card. As outlined in the 8255 Study Card Manual.
6. Study of 8259 Interface Card. As outlined in the 8259 Study Card Manual.
7. Move a block of data from a source address location to target address location using assembly language programming in 8086
8. Design, model and test using HDL (hardware description language), a Sign magnitude adder, BCD incrementor, Gray Counter and LFSR based random number generator.
9. Design model and test using HDL a PWM based LED Dimmer
10. Design model and test using HDL a single switch and keypad matrix de-bouncing system.
11. Design and model using HDL, the following LED multiplexing schemes: regular LED multiplexing, Charlieplexed LED multiplexing.
12. Design, model and test using HDL, a multi-stage Dice game.
13. HDL behavioral model for a 32-bit MIPS ALU

Note: Course teachers may design 3-4 new experiments/small projects in addition to the above suggested practical exercises.

4.3.5 SYLLABI OF PROGRAM CORE COURSES : V SEMESTER

4.3.6 SYLLABI OF PROGRAM CORE COURSES : VI SEMESTER

4.4 DEPARTMENT ELECTIVE COURSES

4.4.1 LIST OF DEPARTMENT ELECTIVES

4.4.2 SYLLABI OF DEPARTMENT ELECTIVES COURSES : V SEMESTER

4.4.3 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VI SEMESTER

4.4.4 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VII & VIII SEMESTERS

4.5 OPEN ELECTIVE COURSES

4.5.1 LIST OF OPEN ELECTIVES

4.5.2 SYLLABI OF OPEN ELECTIVE COURSES : SCIENCES

4.5.3 SYLLABI OF OPEN ELECTIVE COURSES : HUMANITIES & SOCIAL SCIENCES

4.5.4 SYLLABI OF OPEN ELECTIVE COURSES : MANAGEMENT